

TOPICAL REVIEW

A Review of Real-Time Single Image Dehazing Architectures for Embedded Vision Under Adverse Weather Conditions

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ABSTRACT Single image dehazing plays a pivotal role in enhancing visual quality for computer vision systems operating in outdoor or adverse weather conditions. Applications such as autonomous driving, aerial surveillance, smart monitoring, and environmental analysis rely heavily on clear visibility. Haze, caused by atmospheric light scattering, degrades image contrast, sharpness, and color fidelity, significantly impacting the performance of downstream tasks. This review presents a comprehensive overview of single image dehazing techniques developed over the past decade, bridging prior-based physical-model-based methods with recent learning-based approaches. Prior-based dehazing methods, such as the Dark Channel Prior (DCP), Color Attenuation Prior (CAP), and Retinex theory, are classified as domain-based techniques. These approaches leverage handcrafted assumptions to estimate scene transmission and atmospheric light, offering computational efficiency but often struggling with dense or non-uniform haze scenarios. In contrast, learning-based methods including convolutional encoder-decoder architectures, multi-scale feature fusion, generative adversarial networks (GANs), and emerging transformer-based models achieve superior performance by learning haze-relevant representations directly from data. These models generalize well across synthetic and real-world domains and support end-to-end optimization. We evaluate these methods on standard datasets, using metrics like PSNR and SSIM to assess pixel fidelity and perceptual quality. Furthermore, we analyze the impact of various loss functions on training dynamics and restoration outcomes. A particular emphasis is placed on real-time, hardware-friendly dehazing models for edge deployment, considering power, latency, and memory constraints. Finally, we highlight ongoing challenges including generalization to unseen haze conditions, color distortion, limited real-world data, and training complexity. Future directions include transformer efficiency, unsupervised and semi-supervised learning, multimodal fusion, lightweight network design, and domain adaptation. This review serves as a timely resource for advancing robust and deployable dehazing solutions.

INDEX TERMS Single image dehazing, atmospheric scattering, deep learning, real-time dehazing, multimodal fusion, embedded systems.

I. INTRODUCTION

Hazy conditions significantly degrade image quality by scattering light and reducing contrast, posing substan-

tial challenges for computer vision applications such as autonomous driving, surveillance, and environmental monitoring. This degradation arises from the scattering of light by atmospheric particles, leading to direct attenuation and airlight effects, as described by Koschmieder's theory of horizontal visibility [1]. Consequently, restoring clear

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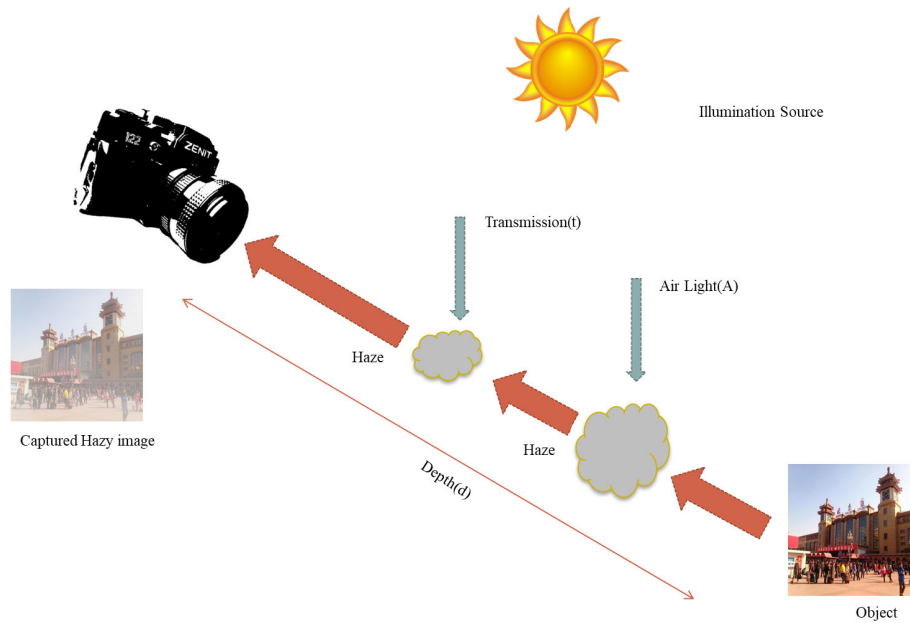


FIGURE 1. Haze formation with Atmospheric scattering model.

images from hazy observations has become a crucial task, driving advancements in the field of single image dehazing. Figure 1 illustrates the scattering of light toward the camera, which contributes to haze formation in captured images. The intensity of haze depends on the transmission map, atmospheric scattering, and scene depth.

Traditional dehazing methods primarily rely on the Atmospheric Scattering Model (ASM), which formulates observed intensity as a function of scene radiance, transmission, and atmospheric light. A pivotal contribution was made by He et al. [2] with the Dark Channel Prior (DCP), an empirical observation that most haze-free images exhibit at least one color channel with minimal intensity in non-sky regions. While DCP performed remarkably under specific conditions, its limitations in handling bright or white objects spurred alternative approaches, such as the color attenuation prior [3] and contrast enhancement methods [4].

With the advent of deep learning, data-driven techniques have outperformed prior-based methods by learning complex mappings directly from data. DehazeNet [5] was among the first convolutional neural networks (CNNs) designed for dehazing, utilizing a trainable framework to estimate transmission maps. Other notable works, like AOD-Net [6], unified atmospheric parameters into a simplified representation, improving reconstruction accuracy. These innovations addressed the limitations of hand-crafted priors and paved the way for more advanced architectures.

Recent developments in network design have significantly enhanced dehazing efficiency and quality. Multi-scale feature fusion networks [7] effectively capture global and local features, while attention mechanisms, such as those in

GFN focus on heavily degraded regions for improved restoration. Lightweight models like FFA-Net [8] prioritize computational efficiency, enabling real-time deployment in resource-constrained environments. Additionally, theoretical advancements such as the Enhanced Atmospheric Scattering Model (EASM) [9] have improved the handling of varying haze densities.

Despite these advancements, challenges remain. A primary limitation is the scarcity of real-world paired datasets, which restricts the generalizability of supervised learning methods. Synthetic datasets, often generated using ASM [6], struggle to replicate real-world haze complexity. Unsupervised and semi-supervised techniques, such as CycleGAN-based approaches [10], attempt to bridge this gap but encounter difficulties in maintaining color consistency and fine detail preservation.

Recently, transformer-based models have emerged as a breakthrough in dehazing, leveraging their ability to model long-range dependencies and global context. Unlike CNNs, which primarily focus on local features, transformers use self-attention mechanisms to establish spatial relationships between distant pixels, effectively handling spatially varying haze. Restormer [11] employs efficient attention mechanisms in the Fourier domain, excelling in high-resolution image restoration. DehazeFormer [12] enhances multi-scale feature extraction, ensuring global consistency and fine detail recovery. Similarly, Uformer [13] integrates hierarchical structures and haze-specific priors, demonstrating superior texture restoration and atmospheric light correction.

In the latest developments, generative adversarial networks (GANs) and diffusion models have gained prominence in

dehazing. Adaptive diffusion-based models [14] refine haze removal by learning hierarchical texture patterns while maintaining natural color distributions. Hybrid transformer-CNN architectures [15] optimize feature fusion to balance computational efficiency and restoration accuracy. Meanwhile, self-supervised contrastive learning [16] has emerged as an effective strategy for training dehazing networks with minimal labeled data, improving generalization across diverse haze conditions.

Furthermore, real-time dehazing solutions have advanced with hardware-efficient designs. Ghosh et al. [17] utilizes quantization techniques to enable efficient deployment on embedded platforms. Additionally, physics-informed deep learning [18] integrates domain knowledge into transformer-based models, improving interpretability and robustness in complex atmospheric conditions.

While these methods achieve impressive results, challenges persist, including high computational demands and reliance on large datasets. Future research aims to optimize these models for real-time applications and address the scarcity of real-world datasets. By integrating transformers, global modeling capabilities with domain-specific knowledge, these approaches represent a promising frontier in single image dehazing.

The progression of single image dehazing methods, as illustrated in Figure 2, highlights a clear evolution from early prior-driven approaches and convolutional networks toward more sophisticated and efficient architectures. Initial models such as DehazeNet [5] and Non-local Image Dehazing [19] laid the foundation with traditional priors and shallow CNNs. This was followed by the emergence of attention-enhanced frameworks like FFA-Net [8] and MSBDN [20], which significantly improved feature refinement. Transformer-based architectures such as GridDehazeNet [21], DehazeFormer [12], and UFormer [13] further advanced the field by capturing long-range dependencies. The integration of generative strategies through GANs, exemplified by Pix2Pix [22], introduced new capabilities in realistic image reconstruction. More recent contributions like UCL-Dehaze [23] and DEA-Net [24] introduced unsupervised learning and edge-aware designs to better handle real-world scenarios. The latest trends, including FusionMamba [25] and multimodal approaches [26], reflect a decisive move toward lightweight, real-time systems capable of robust dehazing in diverse environments.

A. SCOPE AND EXCLUSIONS

This review has special interest in single image dehazing techniques where it is noted that the techniques find use in real-time and embedded vision when in adverse weather conditions. The preview is between traditional models that utilize the previous state of network and the latest advanced models, which include convolutional, transformer, and diffusion-based models. In order to keep the scope clear and concise, multi-image, stereo and video

dehazing approaches that capitalize on either a temporal or multi view are clearly out of this survey. Even though these methods do offer significant contributions to the progressive consistency and motion-sensitive restoration, the algorithmic organization, data requirements, and evaluation metrics of these techniques differ significantly with that of those of single image formulations. Similarly, it is not within the scope of this review to discuss fields such as underwater image enhancement, low light image restoration, and LIDAR or multimodal sensor visibility enhancement. This categorization guarantees a qualitative comparison of individual frame, physics based and learning based dehazing methods that are optimized to execute an efficient realization in real-time and hardware efficient processing.

The major contributions of this review article is as follows

- **Detailed Categorization of Dehazing Techniques**

The paper presents a comprehensive classification of single image dehazing methods, spanning from domain-based priors to modern deep learning models, facilitating a clear understanding of their progression and scope.

- **Integration of Traditional Priors with Deep Models**

It bridges handcrafted priors with CNNs, GANs, and transformer-based models, illustrating how prior-based insights are leveraged within data-driven architectures to enhance dehazing robustness and accuracy.

- **Emphasis on Hardware-Aware and Edge-Compatible Solutions**

The review highlights the growing trend of deploying dehazing models on hardware-constrained platforms using quantization, model pruning, and lightweight architectures like Edge-DehazeNet for real-time inference.

- **Survey of Transformer and Diffusion-Based Innovations**

The study covers recent developments in transformer networks and adaptive diffusion models, showcasing their effectiveness in handling complex haze distributions and supporting efficient processing on high-performance GPUs.

- **Comprehensive result Analysis of Datasets, Loss Functions, and Metrics**

We conducted a comprehensive evaluation of state-of-the-art dehazing networks using benchmark datasets such as RESIDE, I-HAZE, O-HAZE, and Dense-HAZE, incorporating both qualitative and quantitative analyses.

The entire structure has been reorganized with foundational concepts (Section II), datasets and loss functions (Section III), architectural developments (Section IV), experimental results and analysis (Section V), Hardware implementation (Section VI), applications in real world case studies (Section VII), challenges and Future directions (Section VIII) and Conclusion (Section IX).

The remaining review article structure is provided in the Figure 3.

2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Non local image dehazing, Dehaze net	AOD-Net	DCPDN, Gatedfusion-net	Pix2Pix Dehazenet, Griddehaze-net	MSBDN, FFA net	Contrastive learning, Light-Dehazenet	E/d network, Image dehazing transformer	DehazeFormer, UFormer	UCL Dehaze, DEA Net	Fusion Mamba, Multimodel

FIGURE 2. Chronological evolution of single image dehazing methods (2016-2025).

Single Image Dehazing								
I. Introduction A. Scope and Exclusions	II. Foundations and Frontiers: Advances in Dehazing and Haze Modeling	III. Datasets, Loss functions and evaluation metrics	IV. Network designs	V. Comparisons	VI. Practical Evaluation and Deployment Considerations	VII. Applications in Real-World Case Studies	VIII. Challenges and Future Directions	IX. Conclusion
A. Domain-Based Approaches 1 Spatial Domain 2 Transform Domain Approaches 3 Variability Domain B. Learning based Approaches 1 Supervised Learning 2 Semi-supervised Learning 3 Unsupervised learning	A. Datasets B. Loss functions C. Performance metrics 1. Subjective Metrics 2. Objective Metrics 3. Perceptual Metrics 4. Task Driven Metrics 5. Color Fidelity and Distortion Metrics	A. Multi-Branch Architectures B. Dual path Architectures C. Transformer Architectures D. Hybrid Architectures E. Two-stage Networks F. Vision Mamba G. Diffusion Based Models	A. Quantitative Analysis B. Qualitative Analysis C. Color Distortion Analysis		A. Hardware Implementation and Deployment Platforms. B. Comparative Performance Analysis of CNN, GAN, Transformer, and Diffusion-Based Dehazing Models. C. Practical Trade-offs in Single Image Dehazing.	A. Autonomous Driving B. Aerial and Remote Sensing C. Smart Surveillance D. Domain Adaptation for Real-World Scenarios	A. Enhancing generalization across domains B. Blind and zero shot dehazing C. Efficient and real-time processing D. Addressing Compound degradations E. Reducing dependence on paired supervision F. Transparency and use interaction G. Dataset bias and Domain adoption H. Open Challenges in Real-World Dehazing I. Physics-Informed and Hybrid Learning Models	

FIGURE 3. Structure of this review paper.

II. FOUNDATIONS AND FRONTIERS: ADVANCES IN DEHAZING AND HAZE MODELING

Researches in single image dehazing were inspired by scattering models of atmospheres and hand-crafted priors, and these early mathematical models serve as the theoretical base of haze image models. These models, such as Atmospheric Scattering Models (ASM) and priors, fusion, and physical constraints, are aimed at the restoration of lightness and achieved through haze transmission maps, atmospheric luminance, or depth estimation. While these methods are interpretable and provide valuable information about haze image formation, they might fail in complex scenarios of real-world haze images, such as non-uniform illumination, heavily fogged areas, and haze with strong color transfer. In the current section, the theoretical base, inspiring learning models, is established, explaining the limitations of hand-crafted priors and facilities.

Image dehazing is a critical task in computer vision aimed at removing haze, fog, or atmospheric distortions from images to restore clarity and visibility. These distortions, caused by the scattering of light by atmospheric particles, significantly degrade image quality, affecting the performance of downstream applications such as autonomous driving, surveillance, and remote sensing. Approaches to image dehazing can be broadly categorized into domain-based approaches and learning-based approaches. Figure 4 shows the classification of various methods used to approach the dehazing task.

A. DOMAIN-BASED APPROACHES

Domain-based approaches utilizing traditional image processing techniques and prior knowledge of the atmospheric scattering model to enhance and restore images. These methods can be further subdivided into spatial domain, transform domain, and variability domain techniques.

1) SPATIAL DOMAIN

Spatial domain-based approaches work directly on pixel intensities and spatial relationships within the image. These methods focus on manipulating the image's pixel values, usually through mathematical models, to reduce haze and recover the underlying scene. They broadly divided in to image enhancement, image fusion, image filtering and depth map estimation.

a: IMAGE ENHANCEMENT BASED TECHNIQUES

Image enhancement methods like Histogram Equalization (HE), Morphological Operations, and Quad-Tree Decomposition are employed to boost visibility in hazy images by improving contrast and detail. Table 2 presents a comparative analysis of fusion-based approaches across different aspects. A comparative analysis of various enhancement-based dehazing approaches is presented in Table 1.

i) Histogram Equalization (HE):

Enhances contrast by redistributing pixel intensities using the cumulative distribution function (CDF):

$$s_k = (L - 1) \cdot C(r_k), \quad C(r_k) = \sum_{j=0}^k p(r_j), \quad p(r_j) = \frac{n_j}{N} \quad (1)$$

Adaptive methods like AHE apply HE locally:

$$s_k = (L - 1) \cdot C_i(r_k) \quad (2)$$

CLAHE clips the histogram at a threshold T_{clip} :

$$s_k = (L - 1) \cdot C_{clip}(r_k) \quad (3)$$

ii) Morphological Operations:

Improve object definition using dilation, erosion, and edge detection. Gradient-based sharpening uses Sobel

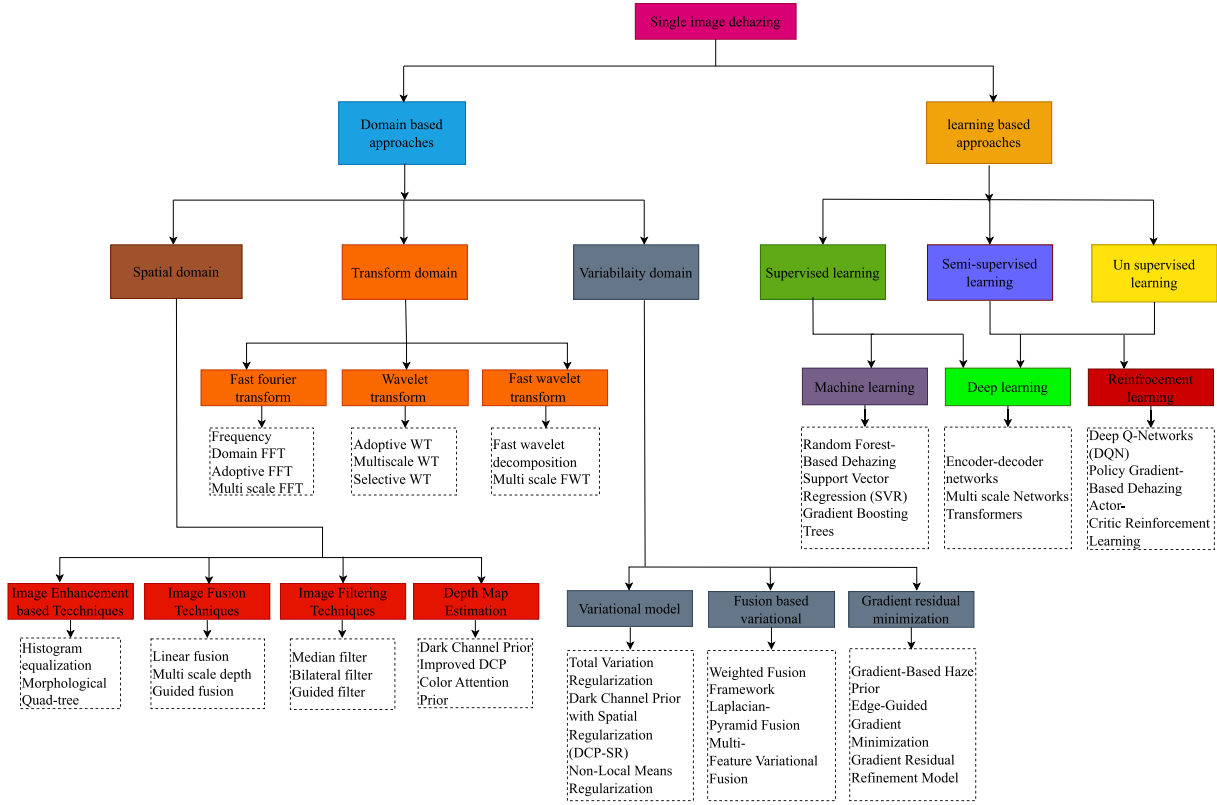


FIGURE 4. Classification of various methods used to approach the dehazing task (Zoom in for better view).

filters:

$$G_x = I * S_x, \quad G_y = I * S_y, \quad G = \sqrt{G_x^2 + G_y^2} \quad (4)$$

Morphological reconstruction is done via iterative dilation:

$$R_k = (R_{k-1} \oplus B) \cap I \quad (5)$$

iii) Quad-Tree Decomposition:

Recursively splits regions based on intensity variance:

$$\text{Var}(R) = \frac{1}{|R|} \sum_{(x,y) \in R} (I(x,y) - \mu_R)^2, \quad \mu_R = \frac{1}{|R|} \sum_{(x,y) \in R} I(x,y) \quad (6)$$

If $\text{Var}(R) > T$, R is divided into subregions. Normalization within each region is:

$$I'_R(x,y) = \frac{I_R(x,y) - \min(I_R)}{\max(I_R) - \min(I_R)} \cdot (L-1) \quad (7)$$

These techniques are often integrated with multi-scale or learning-based models for robust dehazing in real-world scenarios [27], [28], [29], [30].

b: IMAGE FUSION TECHNIQUES

Image fusion techniques combine multiple features or inputs (e.g., transmission maps, depth, or haze-free images) to improve visibility and restore scene structure. Major strategies include Linear Fusion, Multi-Scale Depth Fusion, and Guided Fusion. Table 2 presents a comparative analysis of fusion-based approaches across different aspects.

i) Linear Fusion:

Blends N feature maps $F_i(x, y)$ using spatially varying weights:

$$I_{\text{fused}}(x, y) = \sum_{i=1}^N w_i(x, y) \cdot F_i(x, y), \quad \sum_{i=1}^N w_i(x, y) = 1, \quad w_i \geq 0 \quad (8)$$

Weights can be derived from entropy, contrast, or reliability:

$$w_i(x, y) = \frac{H(F_i)}{\sum_j H(F_j)} \quad (\text{Entropy}) \quad (9)$$

$$w_i(x, y) = \frac{C(F_i)}{\sum_j C(F_j)} \quad (\text{Contrast}) \quad (10)$$

$$w_i(x, y) = \frac{R_i}{\sum_j R_j} \quad (\text{Reliability}) \quad (11)$$

TABLE 1. Comparative analysis of image enhancement-based techniques.

Aspect	Histogram Equalization (HE)	Morphological Operations	Quad-Tree Decomposition
Methodology	Redistributes pixel intensities to improve global contrast.	Manipulates image shapes and sizes using operations like dilation, erosion, opening, and closing.	Divides the image into smaller regions based on intensity variation or haze density, applying adaptive enhancement.
Complexity	Low.	Moderate to high, depending on the operation and region size.	High, due to recursive subdivision and localized processing.
Edge Preservation	Moderate; may cause artifacts like over-enhancement.	High, especially when combined with edge detection techniques.	High, particularly when integrated with contrast enhancement.
Applications	Low-visibility environments such as medical imaging or surveillance.	Edge detection, segmentation, and noise removal in hazy images.	Real-time image enhancement in autonomous driving, remote sensing, and outdoor vision systems.
Computational Cost	Low.	Moderate; dependent on the complexity of operations.	High, due to recursive and adaptive processing.
Real-Time Applicability	High; suitable for quick preprocessing in real-time systems.	Moderate; requires more processing power and time.	Moderate to low; generally needs higher computational resources.

TABLE 2. Comparative analysis of fusion-based dehazing techniques.

Aspect	Linear Fusion	Multi-Scale Depth Fusion	Guided Fusion
Complexity	Low.	Moderate.	Moderate to high.
Performance	Suitable for simple haze patterns.	Excels in scenes with depth variations.	Handles complex textures effectively.
Edge Preservation	Limited.	Moderate.	Excellent.
Applications	Early-stage enhancement methods.	Outdoor scenes with depth variations.	Complex and textured environments.

Due to its simplicity and speed, it's suitable for real-time applications, and can be improved with learned or adaptive weight maps [31], [32].

ii) Multi-Scale Depth Fusion

Uses multi-scale decomposition (e.g., Laplacian pyramids) to estimate depth maps $D^{s_i}(x, y)$ at each scale:

$$D^{s_i}(x, y) = \frac{\beta}{\log \left(\frac{I^{s_i}(x, y)}{J^{s_i}(x, y)} \right)} \quad (12)$$

Final depth map is fused across scales using adaptive weights:

$$D_{\text{fused}}(x, y) = \sum_{i=1}^M w^{s_i}(x, y) \cdot D^{s_i}(x, y) \quad (13)$$

This approach improves both local detail and global structure restoration, particularly in scenes with non-uniform haze [33], [34].

iii) Guided Fusion:

Utilizes guidance maps (e.g., depth, edge, or gradient) to drive the fusion of input features:

$$J(x, y) = \sum_{i=1}^N w_i(x, y, G) \cdot F_i(x, y) \quad (14)$$

where $G(x, y)$ is the guidance map used to compute weights w_i for structure-aware fusion. It helps retain edges and textures, and is effective for joint enhancement under combined degradations (e.g., haze + low-light) [35], [36], [37].

c: IMAGE FILTERING TECHNIQUES

Image filtering plays a crucial role in single image dehazing by reducing noise, preserving edges, and enhancing feature extraction. The comparative analysis of various filtering techniques for image dehazing is illustrated in Table 3.

i) Median Filter

A non-linear filter that replaces each pixel with the median of neighboring values, effectively removing salt-and-pepper noise while preserving edges [38]. It helps maintain structural clarity in hazy images and can be combined with other filters for better results. Adaptive median filters dynamically adjust the window size to better balance noise removal and detail retention [39].

$$J(x, y) = \text{median}\{I(m, n) \mid (m, n) \in W(x, y)\} \quad (15)$$

ii) Bilateral Filter

An edge-preserving smoothing filter that combines spatial and intensity similarity for weighted averaging [40]. It helps retain edges while reducing noise and is defined as:

$$J(x, y) = \frac{1}{W_p} \sum_{(m, n) \in W} I(m, n) \cdot f_s(\|(m, n) - (x, y)\|) \cdot f_r(|I(m, n) - I(x, y)|) \quad (16)$$

where $f_s(\cdot)$ and $f_r(\cdot)$ are Gaussian functions for spatial and range weighting respectively:

$$f_s(d) = \exp \left(-\frac{d^2}{2\sigma_s^2} \right), \quad (17)$$

$$f_r(\Delta I) = \exp \left(-\frac{\Delta I^2}{2\sigma_r^2} \right) \quad (18)$$

and the normalization factor W_p is given by:

$$W_p = \sum_{(m,n) \in W} f_s(|(m,n) - (x,y)|) \cdot f_r(|I(m,n) - I(x,y)|) \quad (19)$$

Hybrid models incorporate bilateral filtering with CNNs for improved dehazing in complex conditions [41].

iii) Guided Filter

Introduced by He et al. [42] this filter uses a guidance image to preserve edges while smoothing. It assumes a linear model between the guidance $G(x, y)$ and the output $J(x, y)$ within a local window ω_i :

$$J(x, y) = a_i G(x, y) + b_i, \quad \forall (x, y) \in \omega_i \quad (20)$$

where a_i and b_i are coefficients computed to minimize the difference between the output and the input. Adaptive guided filters and CNN-guided hybrids further enhance dehazing in the presence of spatially varying haze [43], [44].

TABLE 3. Comparative analysis of filtering techniques.

Aspect	Median Filter	Bilateral Filter	Guided Filter
Complexity	Low.	Moderate.	Low to moderate.
Edge Preservation	Moderate.	High.	High.
Noise Reduction	Effective for small impulsive noise.	Effective for intensity-based noise while preserving edges.	Effective with guidance from structural information.
Applications	Transmission map smoothing and haze removal.	Refining transmission maps and denoising.	Depth map refinement and edge-aware smoothing.

d: DEPTH MAP ESTIMATION

Depth estimation is crucial in image dehazing as it determines object distances from the camera, aiding in haze removal since haze density increases with depth. By analyzing atmospheric light scattering and absorption, depth estimation techniques facilitate accurate haze correction. Common methods include Dark Channel Prior (DCP), Improved Dark Channel Prior (IDCP), and Color Attention Prior (CAP), which extract depth maps from hazy images. Table 4 compares various depth estimation approaches based on key aspects.

i) Dark Channel Prior (DCP):

DCP [2] is based on the observation that in most natural outdoor scenes, at least one color channel has very low intensity in local patches, forming the “dark channel”. For an image $I(x, y)$, the dark channel is computed as:

$$D(x, y) = \min_{c \in \{R, G, B\}} \left(\min_{(x', y') \in \omega(x, y)} I^c(x', y') \right) \quad (21)$$

Using this, the transmission map is estimated under the Atmospheric Scattering Model (ASM) [6]:

$$I(x, y) = J(x, y)t(x, y) + A(1 - t(x, y)) \quad (22)$$

$$t(x, y) = 1 - \omega D(x, y), \quad \text{with } \omega \approx 0.95 \quad (23)$$

The transmission is refined with guided filtering:

$$t_{\text{refined}}(x, y) = \text{GuidedFilter}(t(x, y), I(x, y)) \quad (24)$$

Scene radiance is recovered as:

$$J(x, y) = \frac{I(x, y) - A}{\max(t(x, y), t_0)} + A \quad (25)$$

Recent improvements integrate DCP with deep learning to refine transmission estimation and robustness [45], [46].

ii) Improved DCP (IDCP):

IDCP enhances DCP for dense haze and artificial lighting by refining atmospheric light estimation [3]:

$$A = \max_{(x, y) \in R} I(x, y) \quad (26)$$

Here, R denotes the haze-dominant region rather than using globally brightest pixels. Transmission estimation incorporates a brightness compensation term:

$$t(x, y) = 1 - \omega D(x, y) + \beta S(x, y) \quad (27)$$

with $S(x, y)$ handling bright objects, and β adapting based on local intensity. The haze-free image is then recovered as in DCP. IDCP also benefits from CNN or GAN integration [47], [48].

iii) Color Attention Prior (CAP):

CAP [49] enhances dehazing by using spatial color contrast to estimate transmission:

$$C_{\text{cap}}(x) = \max_{c \in \{R, G, B\}} \left| \frac{\partial I_c(x)}{\partial x} \right| \quad (28)$$

$$t(x) = 1 - \omega C_{\text{cap}}(x) \quad (29)$$

CAP focuses on edge-aware transmission, improving results in areas where DCP fails. Multi-scale models combine CAP across resolutions:

$$t_{ms}(x) = \sum_{s=1}^S w_s \cdot t_s(x) \quad (30)$$

where $t_s(x)$ is the transmission at scale s , and w_s balances local and global contributions. Deep learning further enhances CAP for dynamic and complex scenes [50].

2) TRANSFORM DOMAIN APPROACHES

In the transform domain, techniques focus on transforming the image into a representation where haze can be more easily separated from the underlying scene. This approach improves haze removal by isolating haze effects in a more suitable domain. These methods are broadly classified into Fast Fourier Transform (FFT), Wavelet Transform (WT), and Fast Wavelet Transform (FWT). A comparative analysis of these transform domain methods is presented in Table 5.

TABLE 4. Comparative analysis of depth estimation methods.

Aspect	Dark Channel Prior (DCP)	Improved Dark Channel Prior (IDCP)	Color Attention Prior (CAP)
Methodology	Utilizes the dark channel of the image to estimate transmission	Improves DCP by refining atmospheric light and transmission map estimation	Focuses on color contrast and spatial cues for transmission map estimation
Complexity	Low	Moderate	Moderate
Haze Removal Efficiency	High for simple to moderately hazy images	Better handling of complex or heavy haze conditions	High, especially in challenging scenes with varying haze levels
Edge Preservation	Moderate	High, with better edge recovery in challenging conditions	High, preserves fine details by emphasizing color contrast
Transmission Map Estimation	Relies on dark channel to estimate transmission	Improved estimation, more robust in complex environments	Uses color contrast and spatial cues for refined transmission map
Applications	General dehazing, outdoor scenes with simple haze	Indoor/outdoor scenes, complex haze, artificial lighting	Challenging environments, images with subtle depth variations

TABLE 5. Comparative analysis of transform domain methods.

Aspect	Fast Fourier Transform (FFT)	Wavelet Transform (WT)	Fast Wavelet Transform (FWT)
Key Principle	Converts the image from spatial to frequency domain to separate high and low-frequency components for haze removal.	Decomposes the image into multi-resolution frequency bands, enabling selective enhancement of high-frequency details and suppression of low-frequency haze.	An optimized version of WT that accelerates wavelet decomposition and reconstruction for computational efficiency.
Haze Removal Mechanism	Suppresses low-frequency haze components while retaining high-frequency details like edges and textures.	Targets low-frequency components (representing haze) for removal while enhancing high-frequency features.	Separates haze-related low-frequency components with high computational efficiency, ensuring real-time haze suppression.
Computational Complexity	Moderate, but computationally intensive for large images.	High due to the need for multi-resolution analysis, especially for dense haze removal.	Low, optimized for real-time processing by minimizing redundant calculations.
Adaptability	Can be enhanced with adaptive filtering to handle local variations in haze.	Supports adaptive and multi-scale methods for region-specific haze removal.	Adaptive FWT techniques adjust to local image characteristics, ensuring effective haze removal across varying intensities.
Applications	Outdoor scenes, real-time surveillance, and preprocessing for dehazing networks.	Surveillance, medical imaging, autonomous driving, and detailed image enhancement.	Real-time dehazing in autonomous systems, monitoring, and applications requiring low-latency performance.

a: FAST FOURIER TRANSFORM (FFT) TECHNIQUES

The Fast Fourier Transform (FFT) converts images from the spatial domain to the frequency domain, enabling separation of high-frequency details from low-frequency haze components. Since haze primarily distorts low-frequency regions, FFT helps isolate and suppress these components, thereby restoring contrast and improving overall image clarity.

The Fast Fourier Transform (FFT) converts an image $I(x, y)$ from the spatial domain to the frequency domain:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \quad (31)$$

where $I(x, y)$ is the input image of size $M \times N$.

(u, v) represent frequency components.

$F(u, v)$ is the transformed image in the frequency domain

The inverse Fourier transform reconstructs the spatial image in Eq.32.

$$I(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})} \quad (32)$$

Haze primarily affects low-frequency components of the image due to its smooth and spatially uniform nature. The hazy image $I_h(x, y)$ follows the atmospheric scattering model:

$$I_h(x, y) = J(x, y)t(x, y) + A(1 - t(x, y)) \quad (33)$$

Applying FFT to the hazy image:

$$F_h(u, v) = F_j(u, v)T(u, v) + A \cdot (1 - T(u, v)) \quad (34)$$

where $F_h(u, v)$ is the frequency representation of the hazy image.

$F_j(u, v)$ is the frequency representation of the clear scene.

$T(u, v)$ is the frequency representation of the transmission map.

Recent methods that use FFT-based approaches for dehazing to a large extent enhance the separation of haze and scene content by making use of both frequency and spatial domain cues [51]. They selectively filter out the haze frequencies while letting the fine details through. Moreover, adaptive FFT techniques do the job even better by changing the filters depending on the local image characteristics, thus, allowing for more efficient residue under a variety of conditions.

$$H_{\text{adaptive}}(u, v) = 1 - e^{-\frac{D^2(u, v)}{2\sigma^2(x, y)}} \quad (35)$$

where $\sigma(x, y)$ is space-wise adaptive, changing according to the local image haze density. This technique makes it possible to remove haze in real-time by continually adjusting the filtering operation over the image.

Moreover, multi-scale FFT methods in Eq. 36 have proven effective in handling in dense haze images. By analyzing images across multiple frequency scales, these methods deliver superior results, especially in challenging haze conditions [52]. With their efficient use of the frequency domain, these techniques are highly valuable for real-time applications, such as outdoor photography and surveillance systems.

$$J_{ms}(x, y) = \sum_{s=1}^S w_s \cdot \mathcal{F}^{-1}(H_s(u, v) \cdot F_h(u, v)) \quad (36)$$

b: WAVELET TRANSFORM

The WT provides multi-resolution image analysis by decomposing an image into different frequency bands across multiple scales. This makes it well suited for dehazing, as it separates high-frequency details (edges and textures) from low-frequency components where haze is typically concentrated. By isolating haze-affected low-frequency bands, WT enables effective haze suppression while preserving important structural details.

The WT decomposes an image into multiple frequency bands at different resolutions, given in Eq. 37.

$$I(x, y) = \sum_{j=0}^J \sum_k c_{j,k} \psi_{j,k}(x, y) \quad (37)$$

where: $\psi_{j,k}(x, y)$ are the wavelet basis functions.

$c_{j,k}$ are the wavelet coefficients.

j represents the scale (higher values correspond to lower-frequency components).

k represents the spatial position of the wavelet.

Using Discrete Wavelet Transform (DWT), an image is decomposed into four frequency sub-bands per level shown in Eq. 38.

$$I(x, y) \rightarrow (LL_j, LH_j, HL_j, HH_j) \quad (38)$$

In wavelet-based dehazing, the image is decomposed into a low-frequency sub-band LL_j (haze-dominant) and

high-frequency sub-bands (LH_j, HL_j, HH_j) containing edges and textures. Haze removal is performed by enhancing the approximation component as $LL'_j = \text{Enhance}(LL_j)$, while detail sub-bands are preserved, i.e., $(LH'_j, HL'_j, HH'_j) = (LH_j, HL_j, HH_j)$. The final dehazed image is reconstructed using the inverse wavelet transform: $J(x, y) = \text{IWT}(LL'_j, LH'_j, HL'_j, HH'_j)$.

Recent advancements in WT based dehazing including adaptive wavelet-based methods [53] that apply wavelet decomposition to local patches rather than the entire image. This allows the method to adjust enhancements based on spatial variations in haze density, providing more precise and locally tailored dehazing compared to global wavelet processing.

$$LL_{j'}(x, y) = \text{Enhance}(LL_j(x, y), P(x, y)) \quad (39)$$

where $P(x, y)$ is a local haze measurement, typically derived from the transmission map $t(x, y)$:

$$P(x, y) = 1 - t(x, y) \quad (40)$$

Over the recent year, the wavelet-based dehazing field witnessed over one significant innovation, with the multi-scale wavelet decomposition being one of them that improved the method in terms of dense haze and complex textures performance and quality of the method increased [54]. The application of the Selective wavelet methodologies [55] which combine wavelet attention with multi-scale deep features can be considered a technique that most effectively supports the elimination of the haze. Simultaneously preserves the details, the method known as Discrete wavelet (DWD) [56] was awarded with the physical basis of decomposition. All these approaches result in the positive outcomes of the usage of such likes as surveillance, medical imaging, and autonomous driving. The selective WT is mathematically represented in the Eq. 41.

$$\mathcal{F}_{\text{wavelet}} = \sum_{j=0}^J w_j \cdot \text{WT}_j(I(x, y)) \quad (41)$$

where: w_j are trainable weights learned via CNNs. $\text{WT}_j(I(x, y))$ represents wavelet decomposition at scale j .

The final enhanced image is obtained as an outcome of the inverse wavelet transform. The WT-based dehazing techniques have been very efficient and effective and hence have become a critical tool in image processing in a very diverse range of real-life applications.

c: FAST WAVELET TRANSFORM (FWT)

The FWT is an enhanced form of the non-preconditioned Wavelet Transform, which has higher computational efficiency due to the recursive property of the wavelet decomposition. It makes decomposition and reconstruction fast; it does not require redundancy in calculations and, therefore, it is applicable in real-time execution.

During the dehazing process, FWT particularly differentiates between low-frequency components which haze is most

visible, and high-frequency components which may carry significant features, e.g., edges and textures, and simply removes haze in the low-frequency band while preserving high-frequency information. This efficiency is valuable for real-time tasks such as video surveillance, autonomous navigation, and drone-based imaging.

The FWT decomposes the input image $I(x, y)$ into approximation coefficients A and detail coefficients D (horizontal, vertical, and diagonal components) at multiple levels:

$$I(x, y) \xrightarrow{\text{FWT}} A_j + \sum_{i=1}^j (D_i^H + D_i^V + D_i^D) \quad (42)$$

where j denotes the number of decomposition levels, and D_i^H , D_i^V , and D_i^D represent the horizontal, vertical, and diagonal detail coefficients at level i .

To suppress haze, the low-frequency approximation coefficients A_j are attenuated, and the image is reconstructed using modified coefficients:

$$\hat{I}(x, y) = \text{IFWT}(\alpha A_j + \sum_{i=1}^j (D_i^H + D_i^V + D_i^D)) \quad (43)$$

where $0 \leq \alpha \leq 1$ is a suppression factor applied to the approximation coefficients to reduce the impact of haze.

The latest trends in dehazing by use of the wavelet transform include multi-scale and adaptive filtering. The techniques are dynamic in that they adjust the wavelet basis or filter parameters according to spatial variations in local image properties to deal with the spatially varying haze conditions [57]. Enhanced multi-scale wavelet decomposition improves the performance of dehazing in multi-scale and heterogeneous haze scenarios in the real world scenarios of applications [54].

3) VARIABILITY DOMAIN

These Variability domain-based approaches characterize the local features that describe the variability of haze throughout the image. It also be further divided into Variational Models, Fusion-Based Variational and Gradient Residual Minimization. The comparison of all these methods is provided in Table 6.

α : VARIATIONAL MODEL

Variational approaches aim at achieving the objective of haze removal versus regularization so that the structural integrity is not compromised. It recovers clear images quite effectively through constraints and maintains the edges and the textures so that the dehazing outcomes are more natural and more appealing.

i) Total Variation Regularization

The Total Variation (TV) is a regularization strategy which simply reduces variation within an image to preserve the edges with the aim of smoothing other parts in the image. It is straightforward and efficient, therefore being extensively used in dehazing

particularly when the data of edges is very important. It relies on the ROF model suggested by Rudin et al. [58] that improves the clarity of the structure cleaning the haze in less informative areas without losing the significant details.

The TV-regularized dehazing model seeks to restore a clean image $J(x, y)$ by solving:

$$J^* = \arg \min_J \left[\|J - I\|^2 + \lambda \int_{\Omega} \|\nabla J(x, y)\| dx dy \right] \quad (44)$$

where: $I(x, y)$ is the hazy input image. λ is the regularization parameter controlling smoothness vs. detail preservation.

The first term ensures the dehazed image J remains close to the input.

The second term enforces total variation minimization, removing haze-related smooth variations

ii) Dark Channel Prior with Spatial Regularization (DCP-SR)

DCP-SR builds upon the classical Dark Channel Prior and incorporated spatial regularization to the regularization to capture a more realistic non-uniform haze distribution; encouraging edges to be maintained, as well as the estimation of the transmission map (see Eq. 45), DCP-SR results in superior haze removal performance. Ensuring that the resulting scheme may be effective even in the case of the haze that is hard to model, Li et al. [59] ensured that this solution is one of the reliable ones in the context of actual dehazing applications, particularly in the environmental scenes with the complex structure.

$$t^* = \arg \min_t \left[\sum_{x,y} (t(x, y) - \hat{t}(x, y))^2 + \lambda \sum_{x,y} \sum_{(p,q) \in \mathcal{N}(x,y)} w_{pq} (t(x, y) - t(p, q))^2 \right] \quad (45)$$

Here The second term enforces spatial smoothness, weighted by w_{pq} :

$$w_{pq} = \exp \left(-\frac{(I(x, y) - I(p, q))^2}{\sigma^2} \right) \quad (46)$$

where,

w_{pq} preserves edges by penalizing large intensity differences.

σ controls the sensitivity to intensity differences.

iii) Non-Local Means Regularization

Non-Local Means (NLM) regularization improves dehazing by using similarities between image patches across the entire image, helping preserve fine textures often lost in prior-based methods. Adapted for dehazing by [60], NLM enhances robustness, reduces noise,

TABLE 6. Comparative analysis of variability domain-based approach.

Aspect	Variational Model	Fusion-Based Variational	Gradient Residual Minimization
Key Principle	Formulates the dehazing task as an optimization problem, minimizing an energy function combining fidelity to the hazy image and prior assumptions about the image structure.	Combines multiple dehazed outputs, leveraging variational models to emphasize and merge features like contrast, saturation, and luminance.	Refines gradient information by minimizing haze-induced distortions, ensuring sharper gradients and better edge preservation.
Handling of Variability	Adapts to non-uniform haze patterns using spatially varying parameters and regularization techniques like total variation.	Adapts to local haze variability by weighting different features (contrast, luminance) and emphasizing specific attributes in different regions.	Targets areas with significant haze-induced gradient distortions and applies local dehazing adjustments for variability handling.
Edge Preservation	Strong edge preservation through total variation regularization while smoothing other regions.	Maintains edge details through the fusion of contrast and gradient features, balancing sharpness and smoothness.	Excellent edge preservation by directly addressing gradient distortions and restoring sharp transitions.
Computational Complexity	Moderate to high, depending on the complexity of the energy function and optimization algorithm.	High, as it involves multiple dehazed outputs and feature extraction for fusion.	Moderate to high, as it requires gradient analysis and targeted adjustments.
Performance in Complex Haze	Handles non-uniform haze effectively by incorporating spatial regularization and adaptive parameters.	Performs well in challenging lighting conditions and dense haze by leveraging multiple feature-based dehazing strategies.	Effective in reducing residual haze and recovering clear gradients in images with high haze variability.
Applications	Hazy outdoor scenes, enhancing structural details in natural landscapes, and images with significant haze non-uniformity.	Outdoor scenes with varying lighting, surveillance, and urban environments with diverse haze distributions.	Urban and natural landscapes, edge-critical tasks, and applications requiring sharp gradients and texture recovery.

and maintains visual quality across different regions.

$$\tilde{I}(x, y) = \sum_{(p, q) \in \mathcal{N}} w((x, y), (p, q)) I(p, q) \quad (47)$$

where,

$\tilde{I}(x, y)$ is the denoised intensity at pixel (x, y) .

$I(p, q)$ is the intensity at neighboring pixels (p, q) .

$\mathcal{N}$ is the search window (patch). $w((x, y), (p, q))$ is the similarity weight between patches centered at (x, y) and (p, q) , given by:

$$w((x, y), (p, q)) = \frac{1}{Z(x, y)} \exp\left(-\frac{\|P_{x,y} - P_{p,q}\|^2}{h^2}\right) \quad (48)$$

where.

$P_{x,y}$ and $P_{p,q}$ are patch vectors centered at (x, y) and (p, q) .

$\|P_{x,y} - P_{p,q}\|^2$ is the Euclidean distance between patches.

h is the filter parameter controlling smoothing.

$Z(x, y)$ is a normalization factor ensuring $\sum w = 1$

b: FUSION-BASED VARIATIONAL

Fusion based models can have variational improvement as multiple outputs are merged through amalgamating the strength of different features like contrast, gradient, and saturation. These further generate images of better quality since they balance haze removal with minimizing artifacts hence work well on diverse scenes.

i) Weighted Fusion Framework

The weighted fusion model, a combination of various dehazed results, integrates the results in an adaptive manner with the use of importance maps, which are dependent on the use of texture, brightness, and sharpness. This way, the scheme was able to give a trade-off haze removal and reduced artifacts [61]. The obtained final fused image is obtained with the help of the equation-Eq. 49.

$$J_f(x, y) = \sum_k w_k(x, y) J_k(x, y) \quad (49)$$

where $J_k(x, y)$ represents the k^{th} dehazed result, and $w_k(x, y)$ denotes the adaptive weight maps, satisfying $\sum_k w_k(x, y) = 1$.

ii) Laplacian-Pyramid Fusion

The Laplacian-pyramid fusion employs multi-scale Laplacian to extract and combine detail of various outputs of dehazing, therefore improving the clarity and reducing artifact to sharper results [62]. Rather, it successfully promotes the global framework as well as the local textures.

iii) Multi-Feature Variational Fusion

Multi-feature variational fusion adds saturation, contrast and gradient cues in variational priors to deal with outdoor scenes involving areas of arbitrary haze. Multi-channel solid-state imaging that is enhanced by the sophisticated image processing and object detection algorithms has also demonstrated its enhanced

robustness and image quality that make it fully acceptable in complex outdoor environments by Singh et al. [63] in their study.

c: GRADIENT RESIDUAL MINIMIZATION

Gradient residual minimization pays emphasis on the reconstruction of gradient structure of hazy images to achieve an effective haze removal at the cost of maintaining edge information. These models aim at minimizing the distortions caused by the haze on the gradient structures to improve sharpness and preservation of texture.

i) Gradient-Based Haze Prior (GHP)

Gradient-Based Haze Prior (GHP) is a variation on the previous techniques to eliminate haziness although it is specifically more focused on estimating the gradient distortion caused by haze especially in the vicinity of extremely conspicuous edges or object boundaries. Meng et al. [64] came up with the Gradient-Based Haze Prior.

ii) Edge-Guided Gradient Minimization

The method makes use of strong edges to the maximum as it eliminates gradient variations brought about by haze. This method is effective since it uses edge detecting in the haze removal process thus, strong edges are not lost. Singh et al. [63] have demonstrated that the technique is helpful in enhancing the edge clarity.

iii) Gradient Residual Refinement Model

The Gradient Residual refinement Model minimizes the residual gradients generated by haze that enhances the sharpness of the image and preserves textures. The Gradient Residual Refinement Model is best in the setting where haze is varying and this is one of the issues that the other priors have. Gradients were refined, which enhanced the quality of the dehazed image according to Yang et al. [65] recommendation.

B. LEARNING BASED APPROACHES

Learning based approaches rely on machine learning and deep learning models that are trained on complex haze patterns to then remove the haze in the images using the models. The methods based on learning are classified as supervised, semi-supervised, and unsupervised methods depending on the way they utilize the labeled images. Table 7 explains the learning paradigms.

1) SUPERVISED LEARNING

Supervised dehazing methods train ML and deep models using paired hazy-clear images. Traditional techniques include regression [27], dictionary learning, SVMs [66], Gradient Boosting Trees [67], KNN [68], Bayesian Learning [69], and MRF-based models [70] to predict dehazing parameters.

Deep supervised models directly map hazy to clear images using large labeled datasets. DehazeNet [5] estimates transmission via CNNs, whereas AOD-Net [6] performs

end-to-end dehazing without explicit transmission estimation. Datasets such as RESIDE [71], O-Haze [72], and I-Haze [73] support training, but collecting paired data remains challenging. Techniques like data augmentation [74] and synthetic data generation using GANs [75] address this limitation.

2) SEMI-SUPERVISED LEARNING

Semi-supervised dehazing reduces reliance on large paired datasets by using both labeled and unlabeled images, balancing supervised accuracy with unsupervised flexibility. Domain adaptation bridges the synthetic-real gap. Qiao et al. [76] used adversarial learning for synthetic-to-real adaptation, while Luo et al. [77] applied CycleGAN for domain translation. Su et al. [78] proposed DNMDPT, a semi-supervised model for real-world dehazing.

Self-supervised tasks further improve feature learning. Liang et al. [79] used contrastive learning to distinguish haze levels, and Wu et al. [80] introduced pseudo-labeling to refine predictions. Consistency regularization adds stability. Tian et al. [81] employed a teacher-student framework, improving robustness to varying haze densities.

3) UNSUPERVISED LEARNING

Unsupervised methods remove haze without paired images, suitable when ground truth is unavailable. They often use cycle-consistency, generative models, or self-supervised cues. Li et al. [82] proposed YOLY, a self-supervised network that decomposes hazy images into radiance, transmission, and atmospheric light. Chen et al. [83] introduced Contrastive Disentanglement Learning, and Lan et al. [84] introduced an unpaired diffusion prior to improve real-world adaptation.

Wang et al. [23] developed UCL-Dehaze using contrastive learning with unpaired data. IC-Dehazing [85] incorporated Retinex theory for illumination-controllable restoration. Lan et al. [86] proposed DehazeSB, an optimal-transport-based method linking hazy and clear domains.

These advances improve adaptability and performance of unsupervised dehazing across diverse environments.

a: MACHINE LEARNING BASED METHODS

Machine learning-based dehazing methods utilize statistical models and engineered features, usually manually, to estimate the depth of the scene, transmission maps, and the light of the atmosphere. These types of models are trained on labeled databases and are used to predict clean images using a combination of predictors (contrast, color, edge) to predict clean images. The models include the Random Forest Regression, Support Vector Regression (SVR), Gradient Boosting Trees (GBT), and Markov Random Field (MRF). The models are understandable, effective, yet poor at forecasting hazy images that have complex haze patterns than the deep learning technologies. These methods are compared in Table 8.

TABLE 7. Comparative analysis of learning-based dehazing approaches.

Approach	Supervised Learning	Semi-Supervised Learning	Unsupervised Learning
Machine Learning			
Characteristics	- Requires paired hazy and clear images. - Uses handcrafted features (e.g., Dark Channel Prior, Color Attenuation). - Regression models (e.g., SVM, Random Forest) estimate haze level.	- Uses a mix of labeled and unlabeled images. - Self-training and label propagation techniques help improve learning. - Useful when acquiring paired data is difficult.	- Works without paired images. - Uses clustering or generative models to estimate haze characteristics. - Can be less accurate than supervised methods.
Deep Learning			
Characteristics	- Requires large paired datasets of hazy and clear images. - CNN-based models (e.g., DehazeNet, AOD-Net, GridDehazeNet) learn dehazing mappings. - Multi-scale and attention-based networks enhance results.	- Uses limited labeled data along with unlabeled images. - Self-supervised learning (e.g., contrastive learning, knowledge distillation) helps improve performance. - Adversarial learning (e.g., CycleGAN) can refine dehazed outputs.	- Uses generative models (GANs, VAEs) for haze removal. - Domain adaptation helps transfer knowledge from synthetic to real-world images. - Self-supervised and contrastive learning improve generalization.
Reinforcement Learning			
Characteristics	- The agent is trained with reward functions based on dehazing performance metrics (e.g., PSNR, SSIM). - Policy-based RL models improve dehazing iteratively.	- A hybrid approach where limited supervision is provided. - The agent learns optimal dehazing strategies through exploration.	- The agent explores different dehazing strategies without explicit labels. - Reward functions focus on perceptual quality rather than direct supervision.

TABLE 8. Comparative analysis of variability machine learning-based approach.

Aspect	Random Forest-Based Dehazing	Markov Random Field (MRF)-Based Dehazing	Support Vector Regression (SVR) Dehazing
Key Principle	Uses an ensemble of decision trees to learn a mapping between haze-related features and clear image properties.	Models image pixels as nodes in a probabilistic graphical model, enforcing spatial smoothness in transmission estimation.	Uses regression-based learning to predict transmission maps while minimizing errors within a threshold.
Edge Preservation	Can capture sharp transitions if trained on suitable edge-aware features (e.g., contrast, DCP).	Ensures edge preservation by modeling neighboring pixel relationships while enforcing smoothness constraints.	Maintains smooth transitions in estimated transmission maps, reducing artifacts in the dehazed image.
Computational Complexity	High due to the need for training multiple decision trees and extracting complex features.	Moderate to high, depending on the complexity of the graphical model and optimization techniques used.	Moderate, as it involves solving a convex optimization problem, but can scale with kernel functions.
Performance in Complex Haze	Performs well with diverse haze distributions but may struggle with extreme cases if training data lacks sufficient variability.	Effectively handles haze by enforcing global consistency but may introduce over-smoothing in regions with sharp transitions.	Provides smooth and continuous haze removal but may struggle with highly complex haze structures.

i) Random Forest Regression (RFR)

Random Forest Regression (RFR), proposed by Liu et al. [67], uses an ensemble of decision trees to improve prediction and reduce overfitting. In dehazing, RF estimates the transmission map by learning the relationship between haze features and clear image properties. Tang et al. [87] used handcrafted features like DCP, contrast, and hue disparity to train the model, which was then applied to the atmospheric scattering model for haze removal.

Given a set of training images $T(X) = \frac{1}{M} \sum_{m=1}^M T_m(X)$, RF learns a mapping function as:

$$T(X) = \frac{1}{M} \sum_{m=1}^M T_m(X) \quad (50)$$

where $T_m(X)$ is the prediction from the m^{th} tree and M is the total number of trees.

ii) Support Vector Regression (SVR)

Support Vector Regression (SVR) minimizes prediction errors within a threshold ϵ , producing smooth

transmission maps. Ren et al. [66] used SVR to learn the mapping from haze-related features to clear image properties, effectively estimating transmission maps and reducing artifacts.

For a training set $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^N$, SVR learns $f(X)$ by minimizing:

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \max(0, |Y_i - f(X_i)| - \epsilon) \quad (51)$$

iii) Markov Random Field (MRF)

Markov Random Field (MRF)-based approaches represent image pixels as nodes in a probabilistic graphical model, ensuring spatial smoothness in transmission estimation.

In 2013, Caraffa et al. [70] applied Markov Random Field (MRF) models to dehazing, enforcing similarity among neighboring pixels while preserving sharp edges. This method effectively reduced noise and improved the overall clarity of dehazed images.

The transmission map T is estimated by minimizing the following energy function:

$$E(T) = \sum_i \phi(T_i, X_i) + \lambda \sum_{(i,j) \in \mathcal{N}} \psi(T_i, T_j) \quad (52)$$

where: $\phi(T_i, X_i)$ is the data term measuring how well T_i explains X_i ,

$\psi(T_i, T_j)$ is a smoothness constraint,

λ is a regularization parameter.

b: DEEP LEARNING BASED METHODS

Deep learning methods utilized neural networks to automatically extract features, offering superior generalization to varying haze conditions compared to machine learning techniques. These approaches are mainly categorized into encoder-decoder networks, multi-branch architectures, and transformer-based models illustrated in Table 9.

i) Encoder Decoder networks

The encoder-decoder models are a critical part of image dehazing that learns hierarchical representations to effectively eliminate haze. The new trends saw the introduction of improvements on the old models with improved approaches. Li et al. [88] proposed the Progressive Encoding-Decoding Network (PEDN) that relied on the principle of feature memory and pyramid fusion in order to advance dehazing. Subsequently, Keshaw et al. [89] presented ReEDNet that uses a Cross Feature Aggregation block to retain the spatial features. Another similar model [90] is a transmission map-based approach, introduced by Tran and Park, who modeled their results after a prior knowledge.

Mathematically, the encoding process can be represented as transforming an input x into a latent feature representation z using an encoder function f_{enc} , parameterized by θ_{enc} :

$$z = f_{\text{enc}}(x; \theta_{\text{enc}}) \quad (53)$$

The decoding process then reconstructs an output $\hat{y}$ from z using the decoder function f_{dec} , parameterized by θ_{dec} :

$$\hat{y} = f_{\text{dec}}(z; \theta_{\text{dec}}) \quad (54)$$

ii) Multi Scale Fusion networks

Multi-scale network is a necessary element of image dehazing since they can learn various scales to meet various conditions of haze. Dong et al. [20] have introduced the MSBDN-DFF, which is a multi-scale network that finetuned the task of haze removal. Wang et al. [91] developed MSD-Net that comprised dilated convolution that used self-adaptive attention to encourage the learning of features using RS images. Raju et al. [92] proposed TSMD-Net, a two-stage network, which applied feature fusion with multi-window self-attention, best optimized on Jetson.

The basic dehazing mathematical representation for Multi scale dehazing networks is expressed in Eq. 55

$$F_{\text{refined}} = \sum_{i=1}^n w_i F_i \quad (55)$$

where: F_{refined} is the final dehazed output,

F_i represents the feature maps extracted at different scales,

w_i are learnable weights ensuring optimal feature blending,

n is the total number of branches operating at different scales.

iii) Transformers

Transformers have emerged as a strong alternative to CNNs in computer vision by leveraging self-attention to model long-range dependencies. Vision Transformers (ViTs) [93] have been adapted for tasks like image dehazing, segmentation, and object detection. Unlike CNNs, which rely on local receptive fields, transformers process global context, improving restoration of haze-obscured details. The self-attention function is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad (56)$$

where Q , K , and V are the query, key, and value matrices, and d_k is the key dimension. Multi-head self-attention (MHSA) allows multiple heads to capture diverse features:

$$\text{MHA}(X) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W^O \quad (57)$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (58)$$

A feedforward network (FFN) further processes the features:

$$\text{FFN}(x) = \text{ReLU}(W_1x + b_1)W_2 + b_2 \quad (59)$$

TABLE 9. Comparative analysis of variability deep learning-based supervised approach.

Aspect	Encoder-Decoder Networks	Multi-Scale Fusion Networks	Transformers
Architecture	Consists of an encoder that extracts features and a decoder that reconstructs the image.	Uses multiple branches operating at different scales for feature fusion.	Uses self-attention mechanisms to model long-range dependencies.
Feature Extraction	Captures hierarchical features through downsampling and upsampling.	Extracts features at multiple scales to handle varying haze densities.	Captures global dependencies via self-attention instead of local convolution.
Handling of Haze Variability	Struggles with extreme haze variations due to fixed receptive fields.	Efficiently processes both dense and light haze by using multi-scale feature maps.	Adapts dynamically to haze density by learning global relationships.
Computational Complexity	Moderate complexity; requires sequential processing for reconstruction.	Higher complexity due to multiple parallel branches but optimized for dehazing tasks.	High complexity due to self-attention mechanisms and large matrix computations.
Edge Preservation	Moderate; often needs additional refinement layers.	High; captures fine details across different scales.	Very high; models global and local structures effectively.

Positional encoding is added to retain spatial information.

Recent transformer-based dehazing models show superior performance. DehazeFormer [12] uses transformer blocks with cross-attention, TransWeather [94] employs a lightweight transformer for adverse weather restoration, and UFormer [13] aggregates local and global features effectively. These networks outperform CNN-based methods by capturing long-range dependencies, adapting to varying haze densities, and preserving fine image details.

c: REINFORCEMENT LEARNING BASED METHODS

Reinforcement Learning (RL) offers a novel approach to image dehazing by framing it as a sequential decision-making problem. Unlike supervised learning, RL-based methods do not require paired hazy and clear images for training. Instead, they learn optimal dehazing policies through interactions with the environment, guided by reward feedback. Typically modeled as a Markov Decision Process (MDP), RL-based dehazing allows the agent (model) to iteratively refine the image by selecting actions that enhance visibility, making it particularly useful for real-world scenarios with limited labeled data.

i) Deep Q-Networks (DQN)

Deep Q-Networks (DQN) utilize Q-learning with deep neural networks [95] to approximate the optimal action-value function, which evaluates the best action to take in a given state. In the context of image dehazing, an agent learns to iteratively modify pixel intensities or image features to maximize a reward function based on visibility metrics.

The Q-learning update rule is given by:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left(r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right) \quad (60)$$

where,

$Q(s_t, a_t)$ is the estimated Q-value for taking action a_t in state s_t .

r_t is the reward for the action taken at time t .

γ is the discount factor (determines the importance of future rewards).

α is the learning rate.

ii) Policy Gradient-Based Dehazing

Policy Gradient (PG) [96] methods optimize the dehazing policy by directly computing the gradient of expected rewards with respect to the policy parameters. These methods are particularly useful for continuous action spaces, making them suitable for progressive dehazing where small adjustments improve the image quality.

The policy gradient theorem states:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R(\tau) \right] \quad (61)$$

where: $J(\theta)$ represents the expected reward of the policy π_{θ} .

$\pi_{\theta}(a_t | s_t)$ is the probability of taking action a_t in state s_t under policy π_{θ} .

$R(\tau)$ is the cumulative reward over trajectory τ .

iii) Actor-Critic (AC) reinforcement learning

Actor-Critic (AC) reinforcement learning combines policy-based and value-based methods to enhance decision-making. It consists of an actor, which selects actions based on a policy $\pi(a|s; \theta)$, and a critic, which evaluates actions using a value function $V(s; w)$. This dual architecture improves learning stability by leveraging both direct policy optimization and value function estimation. The actor's update is based on the advantage function $A(s, a) = Q(s, a) - V(s)$, guiding policy improvement:

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \log \pi(a|s; \theta) A(s, a) \quad (62)$$

where α is the learning rate. The critics update refines value estimates using the Temporal Difference (TD) error:

$$\delta = r + \gamma V(s'; w) - V(s; w) \quad (63)$$

$$w \leftarrow w + \beta \delta \nabla_w V(s; w) \quad (64)$$

where β is the critic's learning rate, r is the reward, and γ is the discount factor.

Recent advancements, such as Springenberg et al. [97], have optimized AC methods for large-scale transformer models, while Zhou et al. [98] improved multi-objective learning efficiency. Koh et al. [99] demonstrated the effectiveness of AC in physics-guided environments, reinforcing its versatility across domains.

The classical and physically motivated dehazing technique in this section have revealed the difficulties where priors are manually designed, which cannot reproduce the actual problems in the atmosphere. The presence of the need of careful design of a dataset, strategy of ground-truth acquisition, and loss function is due to the difficulties mentioned in the previous section and is thus naturally addressed in Section III.

III. DATASETS AND LOSS FUNCTIONS AND EVALUATION METRICS

The key success factors of dehazing on the basis of deep learning are large-scale datasets, accurate ground truths, and well-structured loss functions. The drawbacks of classical techniques have been mentioned in Section II, therefore, it appears essential to use strong data-driven supervision to handle real-life atmospheric variations, depth ambiguity, and complex natural scenes.

Within this discussion, the most in-use datasets of image dehazing are discussed. We will have examples of synthetic datasets such as RESIDE, I-Haze, O-Haze and real ones. We get into the loss functions that ensure the model is on track. These are the L1/L2 loss, perceptual loss, and adversarial loss of the GAN-based models. Among the most important measures of the performance of the dehazing methods that are applied to visual quality, structural similarity, and color fidelity, we discuss PSNR, SSIM, and GMSD.

A. DATASETS

Single image dehazing is a critical task in computer vision, and the creation of high quality datasets with appearance of the haze of different levels, depth complexities, and real-life conditions is becoming urgent. A number of datasets have been developed to facilitate research in this field with each having its differences in regard to the data collection technique, weather and data augmentation techniques taken in each. This section outlines the summary of the significant dehazing datasets, their main features identification, and their use. Dehazing benchmark datasets serve a purpose in the

development of model as well as evaluation across different conditions of haze. The largest of them is the RESIDE [71] dataset, that contains both synthetic and real-world hazy images to be used in a supervised learning task. The dataset has various subsets with some such as RESIDE- β which in turn has the Indoor Training Set (ITS) of 13,000 images and the Outdoor Training Set (OTS) of 70,000 images. The hazy look of the Indoor and Outdoor Training Sets and the clear images were generated using depth information and atmospheric scattering model. Moreover, RESIDE- γ , also known as the Real-World Test Set (RTTS), is a collection of 4,322 real-world unpaired hazy photos in the natural setting, which is appropriate to model test without the use of synthetic data.

Other works such as O-Haze [72] and I-Haze [73] are some other contributions, but each has a distinct feature where they gather the real-world haze in both outdoor and indoor environments, respectively, which guarantee the natural haze patterns. The next model is NH-HAZE [100], which introduces non-homogeneous haze, which makes it hard to model the change of haze intensity throughout an image. Dense-Haze [101] aims at testing the models in extreme haze case to evaluate the models in the low visibility degradation. However, HazeRD [102] is a synthetic dataset that was created with depth-aware transmission maps that can be used to simulate haze in a controlled way. HSTS [103] suggests using a hybrid method when combining the images of real and synthetic ones, in order to make better predictions across various haze levels.

Other related datasets are AirFog [104] whose dataset is simulated to depict the effect of fog-haze with depth information to add variety to training conditions. The next recent addition dataset is a Dehaze-V2 [105] where synthetic and real hazy images are merged into a unified dataset with additional weather conditions and variations in depth to enhance the generalization of models. Figure 4 provides images of various kinds of datasets. These combined can work robust in providing researchers with powerful dehazing models which can work in varying environmental conditions and capabilities of haze.

B. LOSS FUNCTIONS

A loss function is a mathematical function that quantifies the difference between the predicted output of a model and the actual ground truth. It provides a way to measure the model's performance and guides the optimization process by minimizing errors.

In image dehazing, the loss function plays a critical role in ensuring that the reconstructed haze-free image is as close as possible to the real clear image. Table 10 shows various loss functions used to restore the original images. Since dehazing is an ill-posed problem, carefully chosen loss functions help preserve details, restore colors, enhance contrast, and reduce artifacts. Different loss functions focus on various aspects, such as pixel accuracy (L1, L2 loss), perceptual quality

TABLE 10. Comparison of loss functions used for image dehazing.

Loss Function	Description	Mathematical Formulation	Advantages	Limitations
Dark Channel Prior (DCP) Loss [106]	Inspired by the DCP technique, enforces that the dark channel of a dehazed image should be close to a clear image.	$L_{DCP} = \left \min_c \left(\min_{y \in \Omega(x)} I_{pred}^c(y) \right) - 0 \right $	Maintains natural contrast and colors.	Less effective in bright regions.
Atmospheric Loss [107]	Models image degradation using the atmospheric scattering model and penalizes inconsistencies between estimated and actual transmission maps.	$L_{atm} = \ I_{pred} - (J \cdot t + A(1 - t))\ $	Physically interpretable, helps guide learning towards realistic dehazing.	Requires good estimation of transmission map and atmospheric light.
Physical Model Loss [108]	Ensures consistency with the atmospheric scattering model by penalizing deviations in estimated transmission maps or scene radiance.	$L_{phys} = \ I_h - (I_d t + A(1 - t))\ ^2$	Ensures physically plausible dehazing. Works well for domain-specific applications.	Requires accurate estimation of transmission and atmospheric light. Struggles with complex real-world haze.
L1 Loss (MAE) [109]	Measures absolute differences between predicted and ground truth images. Less sensitive to outliers.	$L_{L1} = \frac{1}{N} \sum_{i=1}^N I_{pred}(i) - I_{gt}(i) $	More robust to outliers. Produces less blurry outputs.	Slower convergence, does not penalize large errors significantly.
L2 Loss (MSE) [110]	Measures squared differences between predicted and ground truth images.	$L_{L2} = \frac{1}{N} \sum_{i=1}^N (I_{pred}(i) - I_{gt}(i))^2$	Simple, effective, reduces reconstruction error.	Sensitive to outliers, can lead to over-smoothing.
PSNR Loss [111]	Measures Peak Signal-to-Noise Ratio to optimize dehazing quality.	$L_{PSNR} = -10 \log_{10} \left(\frac{MAX^2}{MSE(I_{pred}, I_{gt})} \right)$	Directly correlates with image quality. Penalizes distortions effectively.	May favor over-smoothed images, not always aligned with human perception.
Perceptual Loss [112]	Uses feature maps from pre-trained networks (e.g., VGG) to measure perceptual differences.	$L_{perc} = \sum_i (\phi_i I_{pred} - \phi_i I_{gt})^2$	Preserves textures and fine details. Aligns better with human perception.	Requires a pre-trained network, computationally expensive.
SSIM Loss [108]	Measures structural similarity between predicted and ground truth images.	$L_{SSIM} = 1 - SSIM(I_{pred}, I_{gt})$	Preserves perceptual quality and structural details.	Computationally expensive, may not work well for extreme haze.
Contrastive Loss [113]	Encourages the model to differentiate between similar and dissimilar pairs in feature space.	$L_{contrast} = \sum_{h, I_d} 1 - \cos(\phi(I_h), \phi(I_d))$	Improves feature representation learning. Enhances model's ability to distinguish features.	Requires careful selection of positive and negative pairs. Computationally intensive.
Edge-Preserving Loss [114]	Preserves edge information by penalizing gradient differences between predicted and ground truth images.	$L_{edge} = \sum_i \nabla I_{pred}(i) - \nabla I_{gt}(i) $	Prevents over-smoothing, enhances details.	May amplify noise, computationally expensive for high-resolution images.
Adversarial Loss (GAN) [115]	Uses a discriminator to generate more realistic dehazed images.	$L_{adv} = \mathbb{E}[\log D(I_{gt})] + \mathbb{E}[\log(1 - D(I_{pred}))]$	Helps learn complex haze distributions. Produces realistic outputs.	Training is unstable, requires careful tuning, may introduce artifacts.
Total Variation (TV) Loss [58]	Encourages spatial smoothness by penalizing pixel intensity differences.	$L_{TV} = \sum_{i,j} (I(i+1,j) - I(i,j) + I(i,j+1) - I(i,j))$	Reduces noise, generates smooth images.	Overuse can lead to excessive smoothing.
Charbonnier Loss [116]	A differentiable variant of L1 loss that stabilizes training by preventing gradient discontinuities.	$L_{Char} = \sum_{i=1}^N \sqrt{(I_{pred}(i) - I_{gt}(i))^2 + \epsilon^2}$	Stable training, less sensitive to outliers, preserves details.	Requires tuning ϵ , may still cause slight over-smoothing.

TABLE 11. Image quality evaluation metrics.

Metric	Type	Description	Formula (if applicable)	Range
PSNR (Peak Signal-to-Noise Ratio)	Reference	Measures the ratio of maximum possible signal power to noise power.	$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$	$[0, \infty]$ (Higher is better)
MSE (Mean Squared Error)	Reference	Measures average of squared differences between original and processed images.	$MSE = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n [I(i,j) - K(i,j)]^2$	$[0, \infty]$ (Lower is better)
SSIM (Structural Similarity Index)	Reference	Evaluates similarity based on luminance, contrast, and structure.	$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$	$[-1, 1]$ (Higher is better)
FSIM (Feature Similarity Index)	Reference	Compares phase congruency and gradient magnitude for feature similarity.	$FSIM = \frac{\sum_{x \in \Omega} S_L(x) \cdot PC_m(x)}{\sum_{x \in \Omega} PC_m(x)}$	$[0, 1]$ (Higher is better)
GMSD (Gradient Magnitude Similarity Deviation)	Reference	Uses gradient maps to assess local structural similarity.	$GMSD(x) = \frac{2G_r(x)G_d(x) + c}{G_r^2(x) + G_d^2(x) + c}$	$[0, \infty]$ (Lower is better)
VSI (Visual Saliency-Induced Index)	Reference	Considers human visual saliency and color/structure similarity.	$VSI = \frac{\sum_{x \in \Omega} S_{VSI}(x) \cdot \max(S_r(x), S_d(x))}{\sum_{x \in \Omega} \max(S_r(x), S_d(x))}$	$[0, 1]$ (Higher is better)
AMBE (Absolute Mean Brightness Error)	No-Reference	Measures absolute difference in mean brightness.	$AMBE = \mu_{input} - \mu_{output}$	(Lower is better)
NIQE (Natural Image Quality Evaluator)	No-Reference	Measures deviation from natural scene statistics.	$NIQE = \sqrt{(\mu_1 - \mu_2)^T \left(\frac{\Sigma_1 + \Sigma_2}{2} \right)^{-1} (\mu_1 - \mu_2)}$	$[0, \infty]$ (Lower is better)
Entropy	No-Reference	Measures image information content or detail.	$Entropy = - \sum_i p(i) \log_2 p(i)$	$[0, 1]$ (Higher is better)

(SSIM, perceptual loss), edge sharpness (gradient loss), and realism (adversarial loss). A combination of these losses is often used to achieve the best dehazing results.

C. PERFORMANCE METRICS

Performance metrics play a crucial role in assessing the quality of image enhancement methods, such as image dehazing. They enable both objective and subjective evaluation of how accurately the enhanced image reflects the original or how visually pleasing it appears to the human eye. Generally, these metrics are divided into two main types: subjective, objective and Perceptual metrics. In addition to these, Task-Driven Metrics are used to evaluate the effectiveness of enhanced images in improving the performance of high-level vision tasks such as object detection, recognition, and segmentation.

1) SUBJECTIVE METRICS

Subjective evaluation methods rely on human observers to assess image quality based on visual appeal and perceptual

clarity. Typically, participants are asked to rate images using a fixed scoring system. A widely used method is the Mean Opinion Score (MOS), which involves averaging scores given by multiple viewers to obtain an overall quality measure. Despite providing results that align well with human visual perception, these approaches are not practical for large-scale or real-time applications due to their high demand on time and manual effort [117].

2) OBJECTIVE METRICS

Objective metrics are numerical measures obtained through automated algorithms without relying on human input. Such metrics represent a consistent and scalable way for image quality evaluation for different applications. Mostly, objective evaluations based on a type are divided into two categories, namely full-reference and no-reference metrics. Table 11 elaborates each objective evaluation metric.

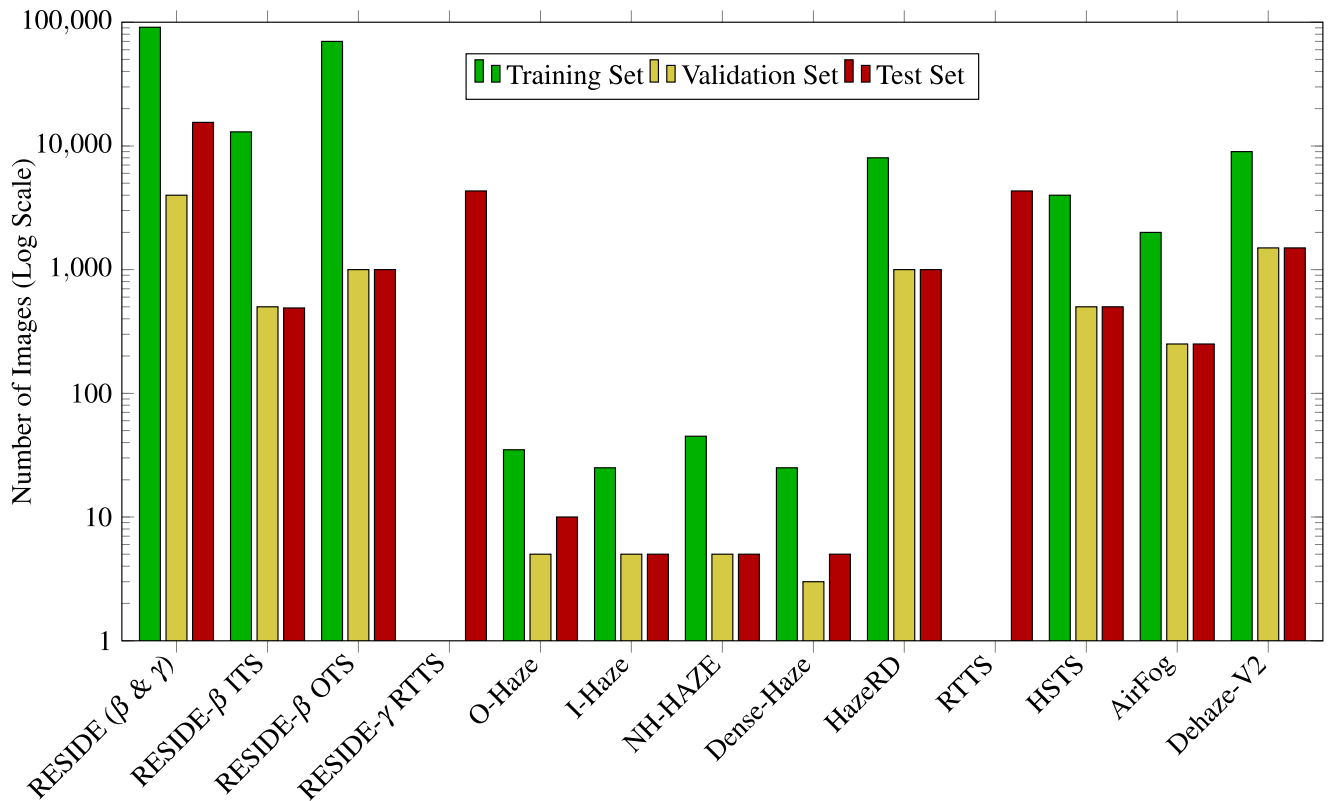


FIGURE 5. Classification of various methods used to approach the dehazing task (Zoom in for better view).

a) Reference-Based (Full-Reference) Metrics

Full-reference image quality measures need a ground truth image to evaluate the restored image quality by matches the enhanced image with original image. These measures are mostly utilized in benchmark datasets and controlled enthrone. The standard full reference metrics are Peak Signal-to-Noise Ratio (PSNR) [111], Mean Squared Error (MSE) [111], Structural Similarity Index Measure (SSIM) [108], Feature Similarity Index (FSIM) [118], Gradient Magnitude Similarity Deviation (GMSD) [119] and Visual Saliency index (VSI) [120]. They all measure the quality of image in different perspectives including pixel precision, structural proximity, semantic importance, and consciousness of visual saliency.

b) No-Reference (NR) Metrics

No-reference (NR) image quality metrics are designed to assess image quality without the need for a ground-truth reference. This makes them particularly valuable for real-world scenarios where clean reference images are unavailable. These metrics estimate quality using various statistical features or perceptual models. Common NR metrics include the Natural Image Quality Evaluator (NIQE) [121], Absolute Mean Brightness Error (AMBE) [122], Perception-based Image Quality Evaluator (PIQE) [123], Entropy [124],

Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE) [125]. NIQE evaluates deviations from natural scene statistics, AMBE quantifies brightness distortions, and Entropy reflects the richness of information within an image.

3) PERCEPTUAL METRICS

Generally, conventional dehazing assessments utilize pixel-level fidelity metrics such as Objective metrics, which do not completely succeed in capturing the perceptual quality or functional usefulness in actual tasks. In order to fill this gap, recent developments have turned their focus to perceptual metrics shown in Table 12, includes Learned Perceptual Image Patch Similarity (LPIPS) [126] and Fréchet Inception Distance (FID) [127].

4) TASK DRIVEN METRICS

In addition to these traditional categories, task-driven metrics illustrated in Table 13 became the important evaluation paradigm, particularly in the practical scenario, where the improved images are for the downstream task. Therefore, task-driven metrics are distinct from fidelity or perceptual quality measures in that they take into account the purpose enhanced images serve to conduct secondary computer-vision tasks like object detection, segmentation,

TABLE 12. Perceptual evaluation metrics for image quality.

Metric	Type	Description	Formula (if applicable)	Range
LPIPS (Learned Perceptual Image Patch Similarity) [127]	Reference	Measures perceptual similarity using deep features from neural networks.	$\sum_l \frac{1}{H_l W_l} \sum_{h,w} \ w_l \odot (F_l(x)_{h,w} - F_l(y)_{h,w})\ _2^2$	[0, 1] (Lower is better)
FID (Fréchet Distance) [128]	Reference	Evaluates perceptual quality by comparing feature distributions of real and generated images.	$\ \mu_r - \mu_g\ _2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$	[0, ∞] (Lower is better)

TABLE 13. Task-driven evaluation metrics for image enhancement.

Metric	Description	Formula (if applicable)	Range
mAP (Mean Average Precision) [129]	Measures detection accuracy by computing average precision over all classes.	$mAP = \frac{1}{C} \sum_{c=1}^C AP_c$	[0, 1] or [0%, 100%] (Higher is better)
Precision [130]	Fraction of correctly detected objects among all detected objects.	$\text{Precision} = \frac{TP}{TP+FP}$	[0, 1] (Higher is better)
Recall [130]	Fraction of correctly detected objects among all ground truth objects.	$\text{Recall} = \frac{TP}{TP+FN}$	[0, 1] (Higher is better)
F1-score [130]	Harmonic mean of Precision and Recall for detection/segmentation tasks.	$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	[0, 1] (Higher is better)
IoU (Intersection over Union) [131]	Evaluates overlap between predicted and ground truth regions in detection or segmentation.	$IoU = \frac{ \text{Prediction} \cap \text{GT} }{ \text{Prediction} \cup \text{GT} }$	[0, 1] (Higher is better)

or classification. On the other hand, mAP [128], recall, precision, F1-score [129] and IoU [130] metrics helps the detector to better recognize objects in haze or low-light environments. These measures are therefore more application focused evaluation metric, which considers real-world utility of enhancement algorithms rather than visual quality. The use of subjective, objective, perceptual, and task-oriented metrics together provides scientists with a complete picture of an image enhancement approach in visual fidelity and performance in conducting downstream tasks.

5) COLOR FIDELITY AND DISTORTION METRICS

Traditional image quality metrics like PSNR and SSIM calculate mainly the structure similarity and brightness tending but they do not consider color fidelity at all. Therefore, it is important when analyzing the perceptual quality of dehazed images to measure the reference image's color tones restoration, saturation, and hue accuracy. The dehazing color metrics depicted in Table 14 represent the widely accepted metrics for color fidelity and distortion.

Among them, the CIE-based measures (CIE76, CIE94, and CIEDE2000) operate in the perceptually uniform CIE Lab color space and provide quantitative estimates of visible color difference, where smaller ΔE values denote better color accuracy. The construction of CIEDE2000 is the most trustworthy, as it considers interactions between lightness, chroma, and hue in a complex manner. Metrics such as ΔE_{HSV} and Angular Error extend this evaluation to the hue-saturation domains, thereby allowing the detection of tiny hue shifts. Meanwhile, the Colorfulness Index (CFI) and Color Histogram Correlation (CHC) evaluate the overall color brightness and the tonal consistency between the dehazed output and the ground truth, respectively. Thus, with the help of these metrics, the color restoration performance of single-image dehazing models is assessed in a comprehensive

and perceptually aligned manner, providing a powerful evaluation framework.

IV. NETWORK DESIGNS

With the realities of the constraints described in Section II and types of datasets and loss functions described in Section III, current dehazing literature has begun to increasingly depend on deep learning architectures. Such models are created and programmed to acquire and discover multi-scale properties, and long-range connections, as well as address the complexity of the atmospheric variations that classical priors or modest supervision are unable to reach.

In this section, we provide an integrated and synthesized analysis of the state-of-the-art methods of deep learning. Based on the major design concepts, we break down the networks, and we demonstrate that the development of data, supervision and feature extraction are inextricably linked to the appearance of stable dehazing models capable of addressing demands of both synthetic and real-world scenarios.

In the last few years, deep learning has had a major impact on the field of image restoration, especially in the area of single image dehazing. Among the distortions caused by haze is the reduction of image visibility which occurs through light scattering and attenuation, making it a very hard challenge to learn the right representations for haze removal. Haze has a multitude of characteristics, which can be as varied from spatial non-uniformity to scene-dependent depth variations, and for that reason, research has been done in different directions, mainly focusing on architectural strategies. Some of the architectural strategies, namely U-Net architectures, dual-path (or two-stream) frameworks, and transformer-based networks, have been the focal point of much interest due to their ability to model intricate feature hierarchies, mask, and capture multi-scale contextual cues,

TABLE 14. Color fidelity and distortion metrics for quantitative evaluation of dehazing.

Metric	Mathematical Expression	Description / Purpose
CIE76 (ΔE^*) [132]	$\Delta E^* = \sqrt{(L_1^* - L_2^*)^2 + (a_1^* - a_2^*)^2 + (b_1^* - b_2^*)^2}$	Measures Euclidean distance between two colors in CIE Lab space; simple but less perceptually uniform.
CIE94 [132]	$\Delta E_{94}^* = \sqrt{\left(\frac{\Delta L^*}{k_L S_L}\right)^2 + \left(\frac{\Delta C^*}{k_C S_C}\right)^2 + \left(\frac{\Delta H^*}{k_H S_H}\right)^2}$	Introduces weighting factors S_L, S_C, S_H and coefficients k_L, k_C, k_H for perceptual uniformity.
CIEDE2000 (ΔE_{00}) [132]	$\Delta E_{00} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2} + R_T \left(\frac{\Delta C'}{k_C S_C}\right) \left(\frac{\Delta H'}{k_H S_H}\right)$	Most perceptually accurate color difference metric; considers lightness, chroma, and hue interactions. Widely used in dehazing evaluation.
Hue-Saturation Difference ΔE_{HSV} [133]	$\Delta E_{HSV} = \sqrt{(H_1 - H_2)^2 + (S_1 - S_2)^2 + (V_1 - V_2)^2}$	Computes difference in hue, saturation, and value; useful for detecting hue shifts and saturation drift.
Colorfulness Index (CFI) [134]	$CFI = \frac{\sqrt{\sigma_{rg}^2 + \sigma_{yb}^2}}{0.5(R + G) - B} + 0.3\sqrt{\mu_{rg}^2 + \mu_{yb}^2}$, where $rg = R - G$, $yb = Y - B$	Quantifies image vividness and color balance. Lower deviation from ground truth indicates faithful reproduction.
Color Histogram Correlation (CHC) [135]	$CHC = \frac{\sum_i (H_i - \bar{H})(G_i - \bar{G})}{\sqrt{\sum_i (H_i - \bar{H})^2 \sum_i (G_i - \bar{G})^2}}$	Measures correlation between RGB (or Lab) histograms of dehazed and reference images; values closer to 1 indicate better color match.
Angular Error (AE) [136]	$AE = \cos^{-1} \left(\frac{\mathbf{c}_1 \cdot \mathbf{c}_2}{\ \mathbf{c}_1\ \ \mathbf{c}_2\ } \right)$	Evaluates angular difference between color vectors; robust to intensity scaling and hue variations.

and global dependencies. All three of these architectures present their own special benefits: U-Net networks are the best at scale-aware processing; dual-path systems are the best when it comes to complementary feature extraction; and self-attention in transformer models is the best for long-range dependency modeling. The upcoming sections go deeper into these architectural categories, outlining their historical development, their support of core principles, and how they have advanced the performance of contemporary dehazing systems.

A. MULTI-BRANCH ARCHITECTURES

Multi-branch networks have been recognized as the most effective way in single-image dehazing since they can accurately and quickly process different scales of image data. The use of the architectures with parallel branches to capture and process the low-, mid-, and high-frequency information comes as a benefit, and thus, non-uniform haze is dealt with better. The initial CNN approaches like AOD-Net [6] failed to adapt to different conditions in contrast, whereas the U-Net versions including MSBDN-DFE [20], Multi-stage net [136], MIDNet [137], and GridDehazeNet [21] could enhance their results with the assistance of dense connectivity and multiscale fusion. The last models such as Dehaze Former- MB are those that combine transformer reasoning with localized haze cues by using attention-guided fusion. Adaptive ones, like CasDyF-Net [138], use cascaded dynamic filters, whereas HAA-Net [139] offers a haze awareness and attention and multiscale frequency boosting combination. Two-stage designs like TSFDNet [140] refine output further by joint processing of haze-density estimation and detail compensation. To sum up, multi-branch architectures are still being developed in the direction of adaptive and context-aware dehazing.

B. DUAL-PATH ARCHITECTURES

Figure 6b represent a major improvement in dehazing outline on Dual-path networks as they are based on global semantics separation and local texture recovery. One such model was DCPDN [141], which estimated the two entities together along two intertwined paths; the Dual Attention Dehazing Network [142] was another model that was applied to enhancing the consistency of the structure by increasing the strength of the connection between the paths; it was among the first models to do it. The modern methods that have subsequently been developed such as CARS-Net [143] allowed the synchronisation of the global context and the spatial detail by using an attention-guided fusion. In addition, the TSMD-Net conceptualized idea enlarged by the Jetson-optimized dual-path pipeline by using a multi-window self-attention and feature fusion, which preconditioned the practical application in the embedded devices.

C. TRANSFORMER ARCHITECTURES

Transformer- based architectures (Figure 6c) have also revolutionized the world of image dehazing, as now long-range dependencies are attainable. Disguised under the ViT [93] model, the restoration model, especially particularly Restormer [11], TransWeather [94], HazelightFormer [144], and Uformer [13] model among others, have chosen to adopt a combination of both global and local attention as a solution to the depth-dependent haze problem. Currently, models like DehazeFormer [12] and AirFormer [145] have already made use of the ideas of windowed attention and dual-branch processing; also, hybrid architectures, like HUFormer [146] and EdgeFormer-Dehaze [147], have already combined the extraction of CNN-based edges with the semantics of transformers, therefore, both being the providers of former performance and the real-time efficiency.

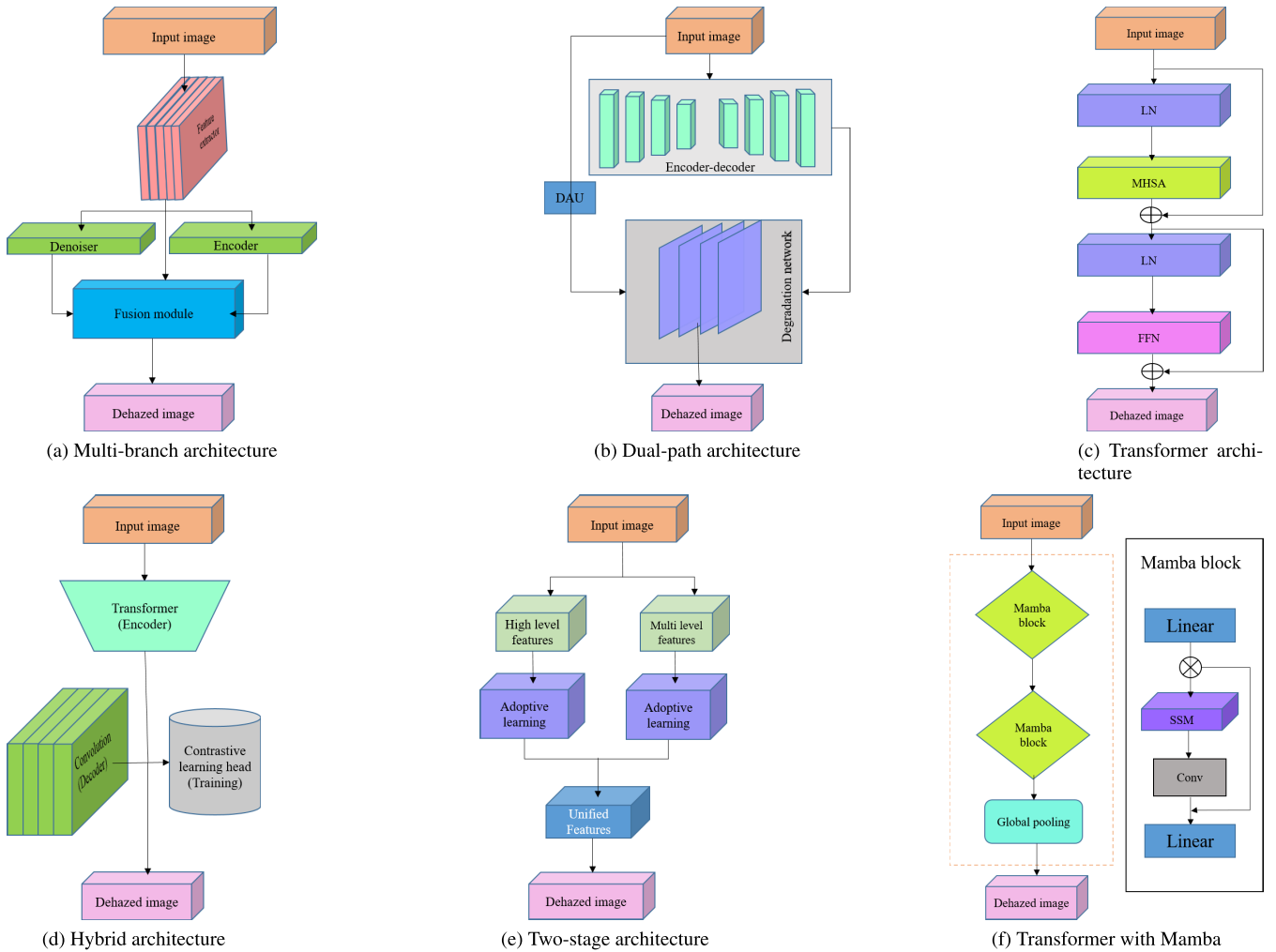


FIGURE 6. Comparison of various single image dehazing architectures. Top row: (a) Multi-branch, (b) Dual-path, (c) Transformer. Bottom row: (d) Hybrid, (e) Two-stage, (f) Transformer with Mamba.

D. HYBRID ARCHITECTURES

Hybrid transformer-convolutional models (Figure 6d) merge local inductive biases with global attention for improved generalization and haze removal. EMST [148] introduces a hybrid attention mechanism for adaptive feature fusion, while DRACO-DehazeNet [149] incorporates contrastive learning and a detail refinement branch to combat texture loss. Frequency-oriented transformers [150] improve reconstruction by operating jointly in the spatial and frequency domains, which is particularly beneficial for remote sensing applications.

E. TWO-STAGE NETWORKS

Two-stage networks (Figure 6e) address dehazing by first performing coarse global correction followed by fine detail refinement. TSNet [151] employs multiscale feature fusion and adaptive weighting for improved restoration under dense haze. Recent methods such as Dehamer [152] and GridDehazeNet++ [153] integrate transformer encoders and

content-adaptive normalization blocks to capture long-range interactions and spatially variant haze distributions, resulting in higher PSNR, SSIM and perceptual quality.

F. VISION MAMBA

Vision Mamba models (Figure 6f) introduce state-space sequence modeling to dehazing, offering efficient long-range feature learning with linear complexity. Mamba blocks combine convolution, state-space modeling, and residual projections [154], providing strong global modeling at lower computational cost compared to transformers [155]. UVM-Net [156] leverages this architecture for real-time dehazing on edge devices, demonstrating robust generalization across diverse haze conditions.

G. DIFFUSION BASED MODEL

The dehazing processing presented in Figure 7 uses a diffusion-based restoration scheme, where a conditional Dehazing Diffusion Probabilistic Model (DDPM) is used, and a transformer-directed reasoning block is used to

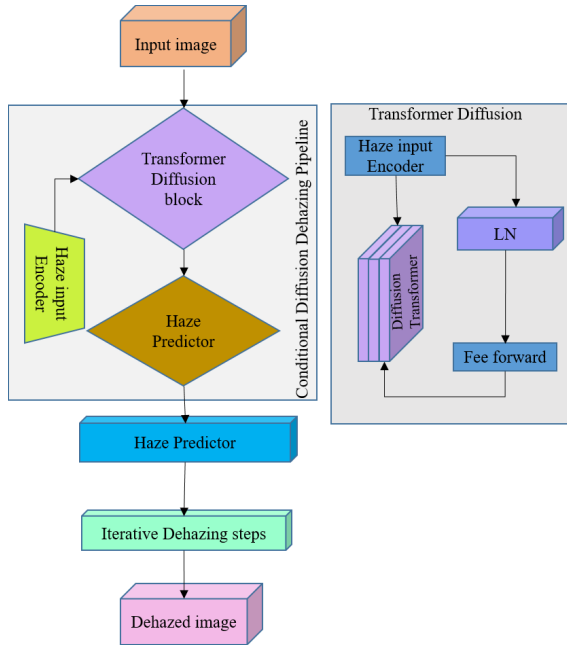


FIGURE 7. Diffusion based model for single image dehazing.

efficiently address the haze-related degradations present in severely contaminated images. It starts with an input hazy image which is initially encoded with a lightweight hazy-image encoder which extracts structural representations and conditioning signals to be used in the following diffusion steps. These coded features are then fed forward to a Diffusion Transformer Block, in which self-attention and cross-attention are used to allow the model to discern long-range dependencies and complex spatial relationships of the scene thus addressing visibility loss and depth ambiguity of haze [11], [12].

The result of this block is fed to a Noise Predictor Block, which predicts the noise associated with a given diffusion timestep, t , as stipulated by the DDPM formulation [157]. The conditioning signal is the noise estimate, which is used on the diffusion trajectory to guarantee the process of reconstruction is stable, since the model uses its predictions based on the actual physical characteristics of haze, and not on solely learned priors. A U-Net/DiT-based noise predictor is then used in the next stage to refine this noise estimate by forecasting the clean image via a series of reversible dehazing autoencoders, decoding at each time-scale $t = T$ all the way to the final time-scale $t = 0$ to successively represent a clean image. The iterative dehazing methods offer soft convergent control over the reconstructed image, and transformer-based conditioning maintains geometric layouts, color distributions and textures that are usually destroyed due to heavy haze.

Finally, the hazy image is recovered via the iterative sampling process with a combination of generative priors of diffusion models and the global context modelling ability of transformer blocks. Recent works evidence that these hybrid transformer-diffusion models can perform better on

demanding outdoor scenarios and natural hazy conditions due to their stability, sample control, and that the images they rebuild are more natural than those produced by the traditional convolutional models [84], [86], [158].

V. COMPARISONS

A. QUANTITATIVE ANALYSIS

Table 15 showcases the evolution of image dehazing methods over the years, highlighting advancements in both performance and efficiency. Figure 8 shows the visual comparison of the SOTA networks on various datasets Like RESIDE, I-Haze, O-Haze and Dense-haze. prior-based methods like DCP (2009) [2] laid foundational work but lacked generalizability across datasets. Early CNN-based approaches such as Dehaze-Net (2016) [5] and AOD-Net (2017) [6] offered lightweight solutions with low computational cost, but their dehazing performance was modest, especially under challenging conditions such as dense haze or in real-world outdoor scenarios.

From 2019 onwards, deep learning models began incorporating attention mechanisms and multi-scale features. GridDehazeNet (2019) [21] introduced grid-based attention refinement, resulting in solid gains across Indoor and Outdoor datasets, though at the cost of very high FLOPs. FFA-Net and MSBDN (2020) [20] improved perceptual quality using dense and feature-fusion strategies, with MSBDN achieving strong balance across all datasets but being significantly heavier in parameters. Meanwhile, transformer-based methods such as DehazeFormer (2023) [12] demonstrated the benefits of global context modeling, achieving excellent results with relatively low parameters and runtime, establishing a new baseline.

Recent innovations in 2024-2025, including TSID-Net [164], UR2P-Dehaze [166], and HITFormer [167], further pushed the envelope. TSID-Net achieved high scores on Indoor benchmarks with only 1.2M parameters and fast inference. UR2P-Dehaze utilized unpaired data with physical priors to outperform earlier models across all datasets with minimal computational load. HITFormer stands out as the current leader, achieving state-of-the-art PSNR and SSIM across every dataset while maintaining a compact model size and runtime $\leq 0.05s$.

Notably, new lightweight contenders like TSMD-Net (2024) [92] aimed at balancing speed, performance, and interpretability by integrating dual-path mixed attention mechanisms. Though it lags behind transformer models in dense haze scenarios, it offers better generalization compared to older CNNs like MSCNN or PAD-Net. Overall, the recent trend clearly favors transformer-based or hybrid designs with attention mechanisms, efficient architecture, and physics-aware modeling for robust, real-time dehazing.

Each row in the image corresponds to a specific dehazing method, and the visual outputs reflect the model's qualitative dehazing performance. By correlating these visual results with the PSNR (Peak Signal-to-Noise Ratio) and SSIM

TABLE 15. Consolidated comparison of SOTA image dehazing methods across multiple datasets.

Method	Year	RESIDE		Indoor		Outdoor		Dense-Haze		Params (M)	FLOPs (G)	Runtime (s)
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM			
DCP [2]	2009	-	-	16.62	0.817	-	-	-	-	-	-	1.100
DehazeNet [5]	2016	21.14	0.847	19.82	0.821	20.33	0.818	-	-	0.008	0.514	0.018
MSCNN [66]	2016	18.48	0.824	15.73	0.744	16.36	0.795	10.22	0.464	2.63	208	0.427
AOD-Net [6]	2017	20.06	0.847	20.51	0.816	19.90	0.816	-	-	0.002	0.005	0.012
GridDehazeNet [21]	2019	23.55	0.915	32.16	0.984	30.71	0.933	19.53	0.726	0.956	452.01	0.070
FFA-Net [8]	2020	22.26	0.932	36.34	0.989	29.03	0.955	22.26	0.781	4.456	287.5	0.060
MSBDN [20]	2020	24.10	0.948	32.77	0.981	28.75	0.948	22.55	0.758	31.35	24.44	0.095
PSD [160]	2021	25.13	0.952	-	-	-	-	-	-	7.1	85	0.080
UCL-Dehaze [161]	2021	23.67	0.939	-	-	-	-	-	-	3.2	64	0.076
FAMED-Net [162]	2021	24.50	0.950	34.10	0.987	-	-	21.52	0.765	2.5	45	0.055
Dehamer [153]	2021	-	-	36.63	0.988	-	-	-	-	132.4	48.93	0.080
SGID-PFF [163]	2021	-	-	38.52	0.991	-	-	-	-	13.87	152.8	0.085
ECFNet [164]	2022	25.79	0.955	35.12	0.988	-	-	23.22	0.794	3.0	60	0.065
DehazeFormer [12]	2023	26.44	0.962	38.93	0.991	30.31	0.958	24.01	0.815	0.686	6.658	0.050
TSID-Net [165]	2024	-	-	39.50	0.993	-	-	-	-	1.2	25	0.048
PAD-net [166]	2024	22.93	0.903	15.97	0.745	18.26	0.702	12.15	0.466	5.8	69	0.208
TSMD-Net [92]	2024	25.31	0.924	22.29	0.849	19.11	0.715	17.47	0.631	16.3	57	0.146
UR2P-Dehaze [167]	2025	27.50	0.970	39.88	0.994	31.12	0.964	25.10	0.837	1.5	30	0.040
HITFormer [168]	2025	28.00	0.975	40.10	0.995	31.35	0.966	25.82	0.843	2.0	35	0.045

(Structural Similarity Index) values in the table, we can draw valuable insights about each model's effectiveness.

B. QUALITATIVE ANALYSIS

The quantitative analysis of the visual performance of various dehazing networks as is depicted in Figure 8 demonstrates that there exist extremely high variations in the ability of the networks to restore color, tone, and manage the remaining artifact. Limited global contextual awareness and the lack of localized transmission estimation that defines the first CNN models such as MSCNN [66] and AOD-Net [6] often result in a slight over-saturation of bright areas. The recent CNNs like GridDehazeNet [21] and FFA-Net [8] have raised the hue stabilizing level by multi-scale fusion and attention, however, traces of reddish or yellowish in the parts of the sky or shadows regions.

Transformer-based models, such as DehazeFormer [12], Dehamer [152], and HITFormer [167], can generate chromatic consistency and smooth tone transition, which is made possible by using long-range dependencies. Besides that, there are other models like TSID-Net [164], PAD-Net [165], and TSMD-Net [92] based on hybrid and dual-path-based methods, that merge spatial attention with multi-scale refinement. In doing so, the resulting image is close to the original image in terms of color and, at the same time, less structural distortion as the transition is also less noticeable. Based on the comparison figures that are shown in Figure 8 we can conclude that transformer-based

and hybrid networks yield superior quality color restoration which is stable and also visually faithful whereas early CNNs even though fast to run, offer uneven tone restoration in the presence of thick haze.

Figure 9 provides a qualitative evaluation of real-scene dehazing performance. Input images (a) are compared with outputs from AOD-Net, DFormer, GridDehazeNet, DHAMER, FFA-Net, MSCNN, TSMD-Net, HITFormer, PAD-Net, and TSID-Net. Newer models such as TSMD-Net (h) and HITFormer (i) recover fine details and natural colors more effectively, whereas earlier methods like AOD-Net (b) and MSCNN (g) frequently retain residual haze or lose contrast, emphasizing the improvements achieved in recent architectures.

The analysis in Figure 10 further compares CNN, Transformer, and Diffusion approaches on Dense-Haze, I-Haze, and O-Haze datasets. AOD-Net (b) partially improves visibility but often leaves light haze and bluish tints. GridDehazeNet (c) produces clearer structures but may introduce slight oversaturation. Transformer models such as LDFormer (d) and TransDehazeNet (e) show superior edge clarity, texture recovery, and uniform color reproduction through global attention mechanisms. Diffusion-based DDPM (f) generates the most natural and realistic outputs via iterative refinement, closely resembling the ground truth (g), though faint haze can remain in extremely dense scenes. Overall, the comparison highlights the progression from early CNN designs toward transformer and diffusion models, with

the latter achieving greater realism, stability, and robustness across diverse haze conditions.

C. COLOR DISTORTION ANALYSIS

Color fidelity has been a major issue in SID for a very long time. In the last few years, there has been a lot of progress in the area of structural and contrast restoration, but many algorithms still show color shifts, desaturation, or unrealistic hue reproduction that negatively affect the quality perceived and the performance of the downstream tasks. Table 16 illustrate the Systematic Comparison of Color Distortion Handling.

1) PRIOR-BASED METHODS

Dark Channel Prior and its variants often suffer from global color shifts and over-saturation, especially in sky or bright regions, due to inaccurate atmospheric light estimation and coarse transmission recovery. Improved approaches such as Enhanced DCP and Color Attenuation Prior introduce brightness and contrast compensation, but their effectiveness remains scene-dependent.

2) LEARNING-BASED METHODS

Deep learning models like DehazeNet and AOD-Net restore global contrast effectively but may introduce color bleeding or unnatural tints because pixel-wise losses prioritize luminance over chromatic accuracy. Perceptual, identity, and chroma-consistency losses reduce these artifacts, while multi-scale and attention-based networks improve local balancing, though localized hue deviations can still occur.

3) TRANSFORMERS AND HYBRID ARCHITECTURES

Transformer models such as DehazeFormer and UFormer, along with hybrid designs, achieve strong global color stability through self-attention and illumination correlation modeling. However, patch-level reconstruction may cause slight local chroma inconsistencies. Physics-guided or chroma-aware loss functions help further minimize these deviations.

4) GENERATIVE AND DIFFUSION FRAMEWORKS

GAN- and diffusion-based models may produce perceptually realistic outputs but sometimes cause color shifts due to adversarial objectives. Incorporating color-preserving cycle losses, identity constraints, or histogram-based matching improves fidelity to true scene colors while maintaining realism.

5) UN/SEMI-SUPERVISED DEHAZING

Unsupervised and semi-supervised approaches face greater color consistency challenges due to the absence of paired data, often resulting in hue shifts, saturation loss, or tone inconsistency. Cycle-consistency, domain adaptation, and contrastive learning help mitigate these issues. Recent self-supervised contrastive frameworks further enhance color stability by embedding color-aware information in the latent

space, though maintaining full chromatic accuracy remains an open research problem.

VI. PRACTICAL EVALUATION AND DEPLOYMENT CONSIDERATIONS

This section describes the practical evaluation and deployment of contemporary single-image dehazing models. Besides the design and benchmark accuracy, the real-world use relies on different factors like the cost of computation, the memory usage, the speed of inference, and the limitations of hardware. So, we review dehazing techniques from a deployment viewpoint, considering the feasibility of the hardware implementation, comparative performance among model families, and the main trade-offs between accuracy, efficiency, and scalability.

A. HARDWARE IMPLEMENTATION AND DEPLOYMENT PLATFORMS

Image dehazing has become an essential component in many real-world computer vision tasks, especially in conditions where visibility is severely degraded due to atmospheric effects like haze and fog. Applications such as autonomous driving, aerial surveillance, remote sensing, and robotic vision critically depend on clear image perception for reliable operation. Although deep learning has introduced powerful dehazing techniques that offer improved accuracy and visual quality, these models typically require high computational power, making them less suited for real-time and edge-based deployments.

To address these limitations, the research community has increasingly focused on hardware-accelerated solutions that can meet the constraints of embedded environments. As shown in Figure 11 platforms such as FPGAs (Field Programmable Gate Arrays), DSPs (Digital Signal Processors), and compact AI computing boards like NVIDIA Jetson and PYNQ-Z2 have gained popularity due to their ability to deliver high-throughput processing with low latency and power consumption. These hardware platforms not only support efficient deployment of deep learning models but also allow prior-based dehazing algorithms to be executed with significant speed-ups, making them ideal for time-sensitive vision applications.

The landscape of image dehazing has transitioned from traditional software approaches to highly optimized hardware implementations. From FPGAs and DSPs to ASICs and AI accelerators like NVIDIA Jetson, the continuous evolution of hardware-aware dehazing systems is enabling real-time, energy-efficient deployment across a wide range of domains. As real-world applications increasingly demand high-performance visual processing under adverse conditions, the integration of specialized hardware with advanced dehazing algorithms will remain a key area of research and innovation.

a) FPGA (Stratix):

The Stratix series from Intel (formerly Altera) represents a class of high-performance FPGAs widely used



FIGURE 8. Comparison of dehazed images of various SOTA networks.

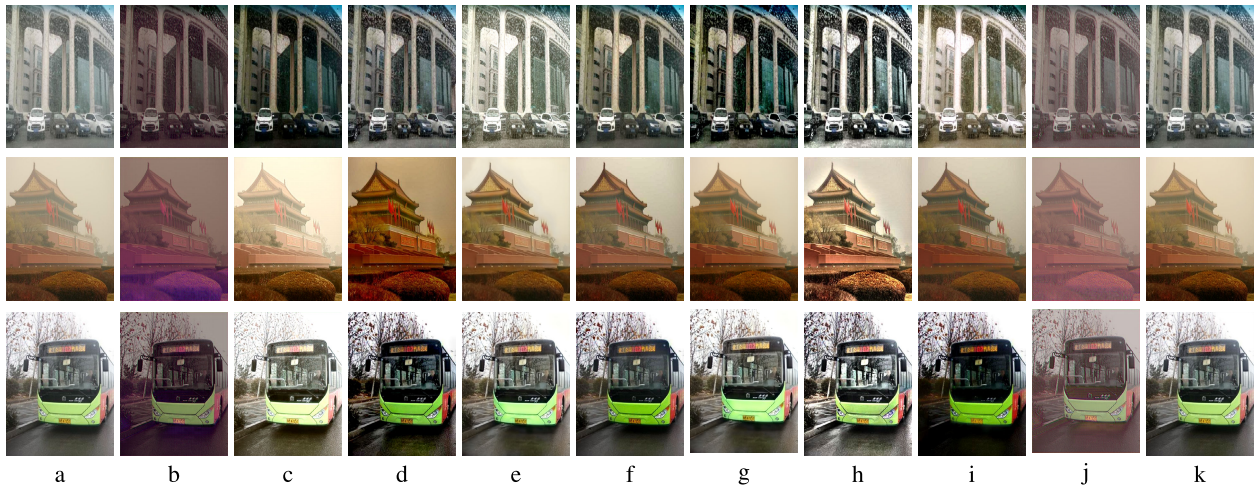


FIGURE 9. Comparison of real-time images on various SOTA networks from left to right: a) input, b) AOD-Net, c) DFormer, d) GridDehazeNet, e) DHAMER, f) FFA-Net, g) MSCNN, h) TSMD-Net, i) HITFormer, j) PAD-Net, k) TSID-Net.

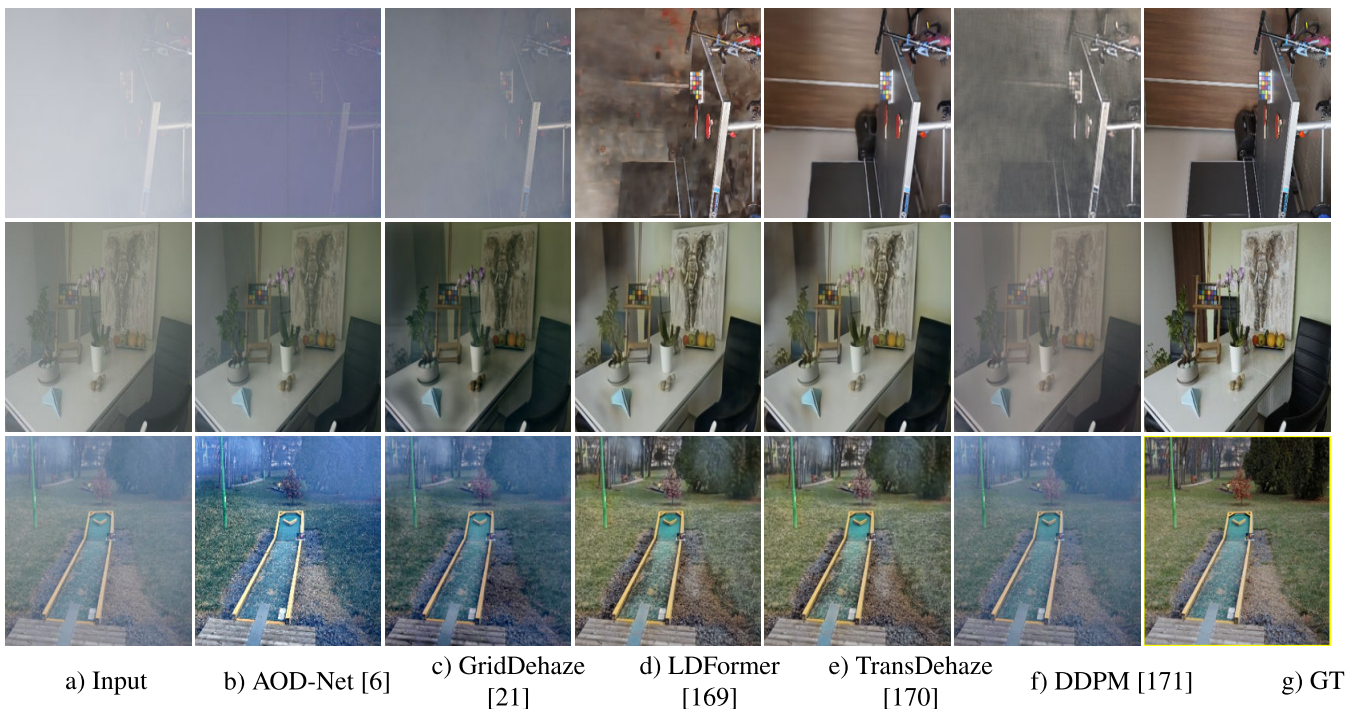


FIGURE 10. Visual comparison of results across CNN (b,c), Transformer (d,e) and Diffusion (f) based methods on Dense-Haze (1st row), I-Haze (2nd row) and O-Haze (3rd row) datasets.

in signal processing and computer vision applications. These devices offer substantial logic density, parallel processing capabilities, and high-speed I/O interfaces, making them suitable for implementing complex algorithms like the Dark Channel Prior (DCP) in real-time. In the context of dehazing, Stratix FPGAs have been utilized to execute resource-intensive operations such as minimum filtering and guided filtering with minimal latency, which is crucial for applications like autonomous driving and live video enhancement.

b) DSP (TMS320C6748):

Texas Instruments' TMS320C6748 is a low-power, high-performance digital signal processor (DSP) from the C6000 family. It is particularly effective for real-time embedded systems where both speed and power efficiency are important. This DSP has been employed in dehazing systems to implement prior-based algorithms like DCP, enabling efficient pixel-wise operations and fast transmission map estimations. Its fixed-point arithmetic capabilities,

TABLE 16. Systematic comparison of color distortion handling across major dehazing approaches.

Approach Family	Typical Color Artifacts	Underlying Cause	Common Mitigation Strategies
Domain-based priors (DCP, CAP, IDCP)	Over-saturation near bright regions; local hue bias (notably in skies and white objects).	Heuristic estimation of transmission and atmospheric light (A); incorrect A leads to global tinting, and guided filtering may amplify errors.	Refined patch-based A estimation, brightness compensation, and post white-balance or color correction mapping.
Supervised CNN encoder-decoder	Over-/under-saturation; color bleeding in fine structures; occasional unnatural tints.	Pixelwise losses ($L1/L2$) emphasize luminance, causing networks to hallucinate chroma for minimizing pixel loss.	Use of perceptual loss, color constancy/chroma loss, and adversarial loss; training with realistic paired datasets.
Multi-scale / attention networks	Improved global color consistency but local tinting when attention mis-focuses.	Attention modules may over-emphasize local textures, neglecting overall chromatic balance.	Hybrid global-local color losses, global color-consistency regularizers, and multi-scale chroma supervision.
Transformer global-context (DehazeFormer, UFormer)	Strong global color preservation; minor local chroma inconsistencies may remain.	Global self-attention models illumination and A effectively, but patch-based reconstruction can distort local chroma.	Physics-based model loss, dedicated color-correction heads, and position/scale-aware chroma modules.
GAN/diffusion/generative models	Realistic yet sometimes inaccurate or patchy colors.	Generators prioritize perceptual realism; discriminators tolerate plausible color deviations.	Color-preserving perceptual (VGG-based) losses, cycle consistency, and identity/color reconstruction constraints.
Unsupervised / semi-supervised learning	Color inconsistency, desaturation, or hue shifts due to domain gap.	Absence of pixel-level supervision; domain adaptation alters color statistics.	Contrastive color alignment, style-transfer regularization, and physical-model consistency constraints.

combined with low power consumption, make it ideal for edge devices in surveillance and mobile robotics.

c) **FPGA (Zynq-706):**

The Xilinx Zynq-706 platform integrates a dual-core ARM Cortex-A9 processor with programmable logic, offering a hybrid architecture for accelerating computer vision workloads. It has been effectively used in hybrid dehazing systems where computational tasks are distributed between the processing system and the programmable logic. For example, the ARM core handles control flow and preprocessing, while the FPGA fabric is used for accelerating pixel-wise computations, such as atmospheric light estimation. This makes it a versatile solution for real-time vision in drones and intelligent machines.

d) **FPGA (PYNQ-Z2):**

The PYNQ-Z2 board, based on the Xilinx Zynq-7020 SoC, is a popular FPGA development platform for embedded vision and deep learning research. It supports Python-based programming (via the PYNQ framework), which simplifies the implementation of neural networks and algorithmic pipelines. In dehazing tasks, the PYNQ-Z2 has been employed to implement CNN-based models like the Dual-Channel Dehaze-Net (DCD-Net), exploiting its parallelism to separately process global and local features for latency reduction in IoT-enabled surveillance applications.

e) **ASIC (65-nm Technology Node):**

Application-Specific Integrated Circuits (ASICs) built on a 65-nanometer technology node offer dedicated,

high-speed performance with very low power consumption. These circuits are custom-fabricated to execute specific algorithms in this case, saturation-based dehazing techniques. Compared to reconfigurable platforms, ASICs achieve superior throughput and energy efficiency, as evidenced by systems delivering 624 megapixels per second at 4K resolution. Such ASIC-based dehazing processors are suitable for deployment in large-scale, ultra-HD video processing applications like smart city monitoring.

f) **Symmetric Multiprocessor System-on-a-Chip (MPSoC):**

An MPSoC integrates multiple processor cores within a single chip, offering massive parallelism and flexibility. These systems are particularly effective for computationally heavy tasks like transmission map estimation in dehazing. By parallelizing the estimation process across different cores, MPSoC-based systems can achieve high-speed dehazing performance suitable for smart infrastructure, autonomous vehicles, and real-time video analytics. Their scalability and power efficiency make them ideal for complex, multi-threaded dehazing pipelines.

g) **FPGA (XC6SLX9):**

The XC6SLX9 is a member of the Xilinx Spartan-6 FPGA family, known for its low cost and efficient logic utilization. Though more constrained than high-end FPGAs, it is capable of handling simplified or optimized dehazing algorithms that require lower resource usage. These devices are well-suited for low-power embedded platforms where basic

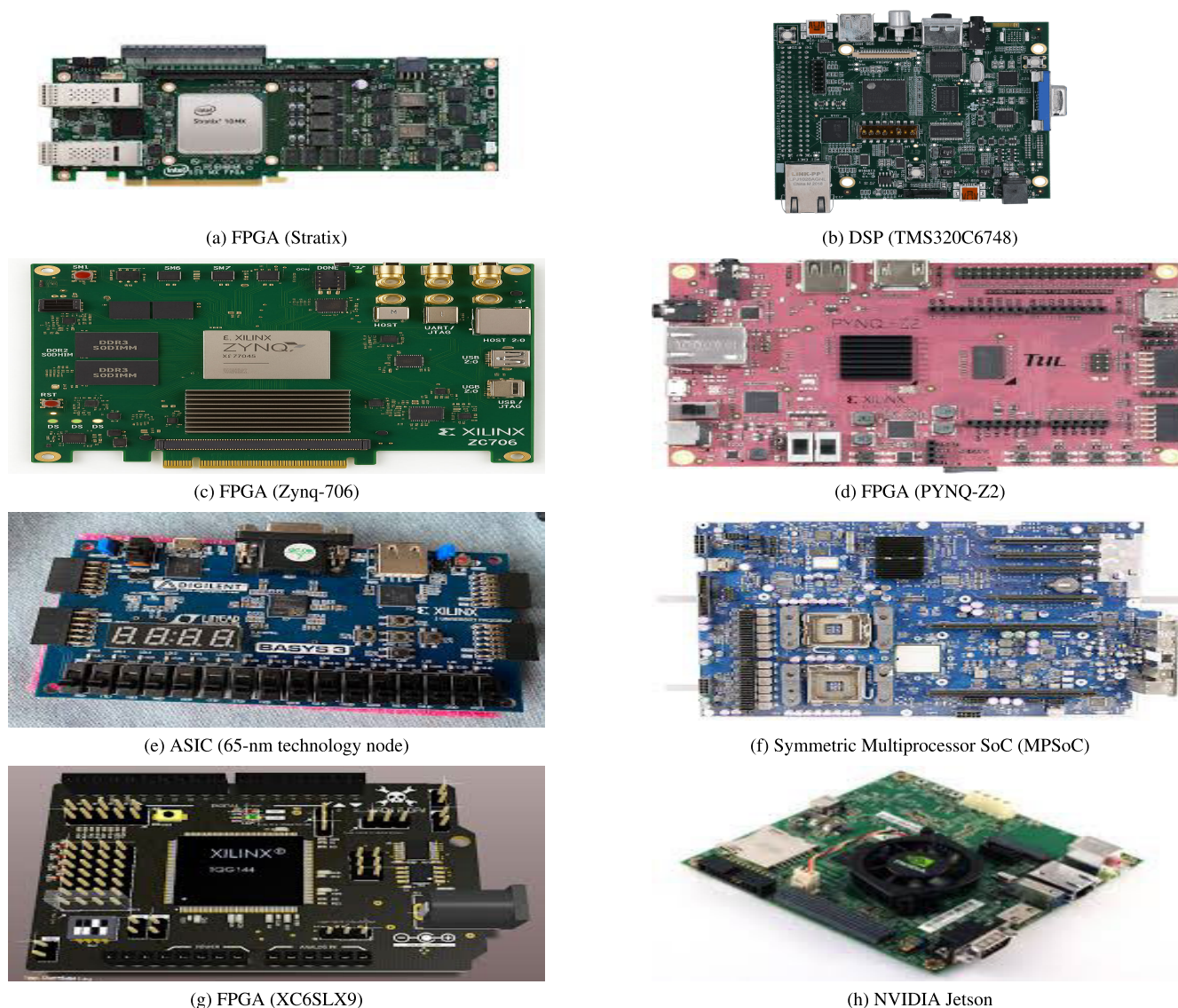


FIGURE 11. Illustration of hardware devices used for implementing dehazing tasks in real time.

dehazing functionality is needed without compromising real-time responsiveness, such as in low-end robotics or portable surveillance units.

h) NVIDIA Jetson:

The NVIDIA Jetson series, such as Jetson Nano, Xavier, and TX2, are powerful embedded AI platforms equipped with CUDA-capable GPUs and ARM-based CPUs. These boards are designed for deep learning inference at the edge and are highly suitable for deploying sophisticated CNN-based dehazing models. For instance, the TSMD-Net (Two-Stage Mixed Dehazing Network) was implemented on a Jetson board using multi-window self-attention mechanisms to enhance restoration quality. Jetson's parallel processing power and deep learning support make it ideal for autonomous drones, self-driving cars, and mobile robots requiring

high-quality visual perception under adverse weather conditions.

Table 17 illustrates the various types of real-time approaches based on hardware implementation. Early efforts in this domain leveraged FPGA implementations of prior-based algorithms like the Dark Channel Prior (DCP). For instance, a 2017 study demonstrated a custom hardware architecture on a Stratix FPGA that executed the DCP algorithm at 116 MHz, offering notable gains in processing speed for high-resolution images [171]. Following this, a 2020 implementation used the TMS320C6748 DSP processor to run a similar algorithm, focusing on energy-efficient processing for power-constrained embedded systems [172]. These works laid the groundwork for a broader shift toward real-time, hardware-based dehazing systems. Over time, the adoption of hybrid architectures, lightweight

CNN accelerators, and parallel processing capabilities in MPSoCs has further advanced the field, enabling high-quality dehazing on compact, efficient, and scalable hardware platforms.

Further innovation led to the development of hybrid hardware architectures that combined the strengths of both FPGAs and DSPs. For instance, a notable 2020 study introduced a dual-platform system utilizing the Zynq-706 FPGA alongside the TMS320C6748 DSP to implement a pixel-wise transmission map estimation algorithm. This configuration improved overall throughput while maintaining low power consumption, positioning the hybrid system as an effective solution for vision tasks in autonomous drones and mobile robotics [173].

As dehazing algorithms became more complex, particularly with the incorporation of perceptual and color consistency constraints, researchers began leveraging advanced design techniques such as Very Large-Scale Integration (VLSI) and Application-Specific Integrated Circuits (ASICs). These custom hardware solutions offer even higher performance and energy efficiency compared to programmable hardware. A 2023 study introduced a saturation-based dehazing algorithm realized on both an FPGA and a 65nm ASIC platform. This implementation achieved a remarkable throughput of 624 megapixels per second at 4K resolution, enabling real-time processing of ultra-high-definition video streams. Such advancements are critical for emerging applications in smart city infrastructure, where real-time processing of high-resolution imagery is required for automated traffic monitoring and infrastructure inspection [174].

Parallel to these efforts, the integration of convolutional neural networks (CNNs) into dehazing pipelines has opened new opportunities for learning-based enhancement. However, running CNNs on conventional CPUs or GPUs is often impractical in embedded contexts due to power and space limitations. To address this, researchers have proposed dedicated CNN accelerators on reconfigurable hardware. In 2022, the Dual-Channel Dehaze-Net (DCD-Net) was implemented on the PYNQ-Z2 FPGA platform. This architecture featured two parallel processing pipelines to handle global and local image features independently, resulting in reduced processing latency and efficient inference for IoT-enabled surveillance systems [175].

The trend continued in 2024 with the introduction of Light-DehazeNet, a lightweight CNN-based dehazing model tailored for FPGA deployment. The network achieved a real-time processing rate of 105 frames per second, while consuming significantly less power, an essential requirement for use cases such as drone imaging and portable vision systems [176]. In the same year, Multiprocessor System-on-a-Chip (MPSoC) architectures emerged as another promising approach. By leveraging multiple processing units, these systems enabled concurrent estimation of transmission maps, substantially accelerating the dehazing pipeline. Their scalable design and real-time capabilities make them highly

relevant for intelligent transport systems and large-scale smart city deployments [177].

More recently, the use of NVIDIA Jetson AI accelerators has gained traction due to their powerful GPU capabilities and compatibility with deep learning frameworks. These platforms allow efficient execution of complex models involving attention mechanisms and multi-scale feature fusion. In 2024, the Two-Stage Mixed Dehazing Network (TSMD-Net) was implemented on a Jetson board, integrating a multi-window self-attention module to capture both local and global haze patterns. This design significantly enhanced image restoration quality while maintaining computational efficiency, rendering it ideal for use in mobile robotics and autonomous platforms [92]. Additionally, a separate study proposed a self-adaptive deep learning-based dehazing system tailored for autonomous vehicles. This model dynamically adjusted its computational load based on environmental conditions and input complexity, delivering high-quality dehazed outputs with minimal latency and power consumption [178].

B. COMPARATIVE PERFORMANCE ANALYSIS OF CNN, GAN, TRANSFORMER, AND DIFFUSION-BASED DEHAZING MODELS

Table 18 shows a comparative analysis among CNN, GAN, transformer, and diffusion models concerning model size, complexity cost, feasibility of computation time, and hazy image restoration capabilities. Among these, methods based on CNN architectures including AOD-Net [6], FFA-Net [8], Dehaze-Net [5], and GridDehazeNet have the least parameter numbers (2-15 million) and FLOPS complexity measures (5-20 billion) and thus have real-time processing capabilities with 30-60 frames per second processing speed using conventional GPUs. Moreover, these architectures have less memory requirements (5-30 MB) and hence are suitable for embedded systems such that they lack capabilities to perform global-level analysis due to relying solely on local context information.

GAN based models such as Cycle-Dehaze [10], DA-GAN, and Pix2Pix derived models [22], [141] make use of adversarial learning for better textured image generation and perceptually improved reconstructions. They have a moderate model size of 20-80M parameters with a higher cost of 40-100 GFLOPs per image, leading to higher memory requirements of 50-150 MB and lower inference speeds of 10-20 FPS. Moreover, mode collapse or hallucinations during adversarial learning can add to their unreliability in real-time processing.

The dehazing models using transformers such as Restormer [11], Uformer [13], DhFormer [179], and AirFormer utilize multiple self-attention in order to establish long-range dependency and global relationships. Although these state-of-the-art models achieve excellent performance, they not only consume higher computation cost with a higher number of GFLOPS (80-200) but also have a larger parameter size in terms of M parameters (25-60M) and higher memory

TABLE 17. Comparison of hardware-based image dehazing implementations.

Year	Category	Title	Hardware Platform(s)	Key Contributions	Citation
2017	FPGA-Based Dehazing	Hardware Implementation for Real-Time Haze Removal	FPGA (Stratix)	Real-time haze removal using dark channel prior, achieving 116 MHz on Stratix FPGA, optimized for large images.	TVLSI 2016
2020	DSP-Based Dehazing	Implementation of Single Image Dehazing System on DSP Processor	DSP (TMS320C6748)	DSP-based dark channel prior dehazing system, optimized for low-power embedded systems.	Springer 2020
2020	FPGA + DSP Hybrid Dehazing	A Novel, Fast and Efficient Image Dehazing Algorithm on Embedded Hardware	DSP (TMS320C6748), FPGA (Zynq-706)	Pixel-wise dehazing algorithm implemented on DSP and FPGA, faster execution without significant quality loss.	Springer 2020
2022	FPGA-Based Dehazing (IoT)	A Dual-Channel Dehaze-Net for Single Image Dehazing in Visual IoT using PYNQ-Z2 Board	FPGA (PYNQ-Z2)	Dual-channel network for single-image dehazing, optimized for low-power IoT devices, achieving real-time processing on PYNQ-Z2.	TASE 2022
2023	FPGA-Based Dehazing	VLSI Design of Saturation-Based Image Dehazing Algorithm	FPGA, ASIC (65-nm)	15×15 minimum filter-based algorithm, processes 4K at 70 fps with 624 Mpixels/s throughput, using 13.2k logic gates.	TVLSI 2023
2024	CNN-Based FPGA Dehazing	High-Speed Hardware Accelerator for Light-DehazeNet	FPGA	Lightweight CNN (Light-DehazeNet) on FPGA, achieving 105 fps, optimized for real-time applications.	Springer 2024
2024	MPSoC-Based Dehazing	A Symmetric Multiprocessor System-on-a-Chip-Based Solution for Real-Time Dehazing	MPSoC	Self-calibrating dehazing algorithm, optimized for balanced performance and efficiency on MP-SoC systems.	MDPI 2024
2024	FPGA-Based Dehazing	Design of an FPGA-Based High-Quality Real-Time Autonomous Dehazing System	FPGA	Self-calibrating dehazing using FPGA, real-time adaptive control for autonomous vision systems.	MDPI 2024
2024	Jetson Board-Based Dehazing	TSMD-Net: Two-Stage Mixed Dehazing Network with Feature Fusion on Jetson Board	NVIDIA Jetson Board	Multi-window self-attention mechanism, optimized for low-power edge AI deployment on Jetson for dehazing in real-time scenarios.	DSP 2024

TABLE 18. Analytical comparison of CNN, GAN, transformer, and diffusion models for single image dehazing.

Aspect	CNN Models	GAN Models	Transformer Models	Diffusion Models
Representative Methods	AOD-Net, FFA-Net, GridDehazeNet	Cycle-Dehaze, DA-GAN, Pix2Pix variants	Restormer, Uformer, DehazeFormer, AirFormer	DDPM-Dehaze, Adaptive Diffusion Dehazing, CDDPM variants
Parameters	2–15M	20–80M	25–60M	100–500M
FLOPs (for 256×256)	5–20 GFLOPs	40–100 GFLOPs	80–200 GFLOPs	500–3000 GFLOPs (50–200 sampling steps)
Inference Speed	30–60 FPS (real-time)	10–20 FPS	10–25 FPS	0.5–2 FPS (slow, iterative sampling)
Training Time	Low–Moderate; converges stably	High; adversarial training unstable	High; requires long epochs, large batch sizes	Very high; noisy score matching with 300+ epochs
Memory Use (VRAM)	Low (5–30 MB)	High (50–150 MB)	Moderate–High (40–100 MB)	Very high (>200 MB)
Strengths	Fast, lightweight, edge-friendly; strong local features	Produces sharp textures, high perceptual realism	Global context modeling; high PSNR/SSIM; strong structure recovery	Best perceptual quality; strong color & texture fidelity; robust to real haze
Weaknesses	Limited long-range reasoning; weak on extremely dense haze	Training instability, mode collapse, hallucinated details	High compute + memory; harder for mobile deployment	Extremely slow; heavy compute; unsuitable for real-time systems
Color Accuracy (ΔE)	Moderate desaturation under heavy haze	Good, but sometimes over-enhanced colors	Mild desaturation in thick haze regions ($\Delta E \approx 2.5$)	Best color accuracy ($\Delta E \approx 1.2$)
Best Use Cases	Real-time systems, mobile devices, robotics, surveillance	Perceptual enhancement for outdoor scenes	High-quality restoration, autonomous driving, aerial imaging	Offline, high-fidelity reconstruction, unpaired real-world haze

usage in terms of VRAM usage in MB (40-100) compared to existing models. Moreover, they show a slower processing speed in terms of FPS with 10-25 FPS compared to CNN models, but these processing speeds can be optimized for real-time processing after careful tuning. Based on these models, transformers in most cases show a slight desaturation effect in very dense hazes.

Based on diffusion, DDPM-Dehaze, CDDPM models, and Adaptive Diffusion Dehazing [14] are the most computationally expensive methods. They have a large parameter scale (100-500M) and a high number of FLOPS (500-3000 GFLOPS) because of the dehazing operations performed in each iteration, which can go up to 50-200. Their network speed is low (0.5-2 FPS) with a requirement of over 200 MB of VRAM. They have been observed to have a better level of perceptual accuracy, texture preservation, and a smaller color error margin ($\Delta E = 1.2$) compared to other methods. In summary, our comparative analysis shows that CNNs are still the most suitable model when it comes to real-time processing in edge and mobile environments because of their excellent efficiency. GANs improve perceptual clarity but are not stable models. Transformers can achieve a balance concerning accuracy and complexity, which make them most suitable for high-quality dehazed image generation. Diffusion models will achieve better restoration accuracy but are not suitable for real-time processing in edge and mobile devices because of their high complexity.

C. PRACTICAL TRADE-OFFS IN SINGLE IMAGE DEHAZING

The evolution of single image dehazing algorithms is inherently driven by trade-offs between accuracy, computational

efficiency, data requirements, and deployment feasibility. An explicit conceptualization of these trade-offs is crucial to guarantee that model design is not just theoretically viable but also operationally in agreement with the limitations and requirements of real-world usage.

1) ACCURACY VS. COMPUTATIONAL COST

CNN-based models reach a good commitment between accuracy and computational efficiency, making them suitable for real-time and incorporated applications (e.g., Dehazenet, AOD-Net, FFA-Net). Transformer-based models like DehazeFormer and UFormer provide better PSNR/SSIM metrics by capturing long range dependencies, but at the cost of higher latency and higher memory demands, which limits their edge implantation. The diffusion models further press the quality of restoration, reaching SOTA perceptual metrics, but suffer from prohibitive inference times (1 to 5 FPS) and excessive use of GPU memory. Therefore, there is an evident trade-off between accuracy and computational efficiency in real-time performance.

2) SUPERVISED VS. UNSUPERVISED LEARNING

Supervised learning is still prevalent because of the robust mapping between haze and haze-free pairs, but it is plagued with overfitting against synthetic distributions and entails the use of expensive paired datasets (e.g., RESIDE). Semi-supervised and unsupervised paradigms, like CycleGAN-based methods and UCL-Dehaze, reduce this reliance through utilizing unpaired data, and hence enhance generalization of the real world scenario. These models, though, tend to generate less stable outputs

and sometimes they trade fine-grained detail at the cost of fully supervised architectures. The trade-off here is in the balance of data availability and restoration accuracy.

3) SYNTHETIC VS. REAL-WORLD DATASETS

Synthetic datasets such as RESIDE give large, diverse training samples and allow for fair benchmarking. However, they do not reflect the full richness of realistic atmospheric scattering and, thus, degrade model performance when models are deployed in field conditions. Real world datasets like I-Haze, O-Haze, and Dense-Haze are more realistic but are small and restrict deep model training. Therefore, models that are trained on synthetic data alone tend to generalize poorly, whereas those dependent on limited real-world data may under fit. Synthetic pre-training with subsequent real world fine tuning or domain adaptation hybrid approaches are becoming increasingly effective solutions to overcome this gap.

These trade-offs point out that no uniquely dominated paradigm of universally dominates; Instead, the choice of method depends on application constraints. For embedded vision, light-weight CNNs and efficient transformer variants continue to be favored, while offline applications can afford diffusion-based or unsupervised frameworks that focus on perceptual fidelity and domain robustness.

VII. APPLICATIONS IN REAL-WORLD CASE STUDIES

Beyond the traditional image quality evaluation, the effectiveness of practical dehazing methods is better reflected under their influence on computer vision applications. It proves that applications for visibility restoration are actually dependent on perception system reliability, and particularly in real-life scenarios, safety-critical and monitorial ones. Case studies have shown that improved visibility contributes significantly to the robustness and accuracy of such systems.

Incorporating dehazing as a preprocessing step has been shown to significantly improve detection, tracking, and recognition performance under adverse weather conditions. Table 19 illustrate Real-world applications of single image dehazing with benchmark improvements from autonomous driving to UAV surveillance, performance benchmarks consistently show that visibility restoration directly boosts detection, recognition, and decision-making accuracy, making dehazing indispensable for safety-critical applications.

A. AUTONOMOUS DRIVING

Hazy conditions significantly impair object detection in autonomous vehicles, where failures can compromise safety. Incorporating dehazing as a preprocessing operation has been demonstrated to yield quantifiable gains. For example, using AOD-Net as a preprocessor to Faster R-CNN increased the accuracy of the vehicle detection by about 12% mAP for synthetic hazes. More recent transformer-based algorithms like DehazeFormer and Restormer further enhanced the

accuracy of YOLO-based detection pipelines to provide stronger perception in poorer weather.

B. AERIAL AND REMOTE SENSING

In satellite imagery and UAV surveillance, haze impedes target identification and terrain mapping. GAN-based approaches such as DCPDN and Pix2Pix have increased the semantic segmentation accuracy, improving the score of the intersection-over-union (IU), which is 8–10% on the haze-degraded aerial dataset. Transformer-based models scale these advantages by recovering fine texture and structural information, which are important for applications such as agricultural monitoring and disaster assessment.

C. SMART SURVEILLANCE

Outdoor surveillance systems tend to work under low-visibility environments. Light-weight CNNs, such as FFA-Net and MSBDN, have been applied to real-time enhancement and are compatible with latency. On the RESIDE-RTTS dataset, Edge-DehazeNet preprocessing enhanced pedestrian detection accuracy by about 9% mAP, highlighting its application in improving safety and security surveillance.

D. DOMAIN ADAPTATION FOR REAL-WORLD SCENARIOS

Although models are trained mostly on synthetic datasets, performance is often compromised during deployment to real-world outdoor haze. Semi-supervised and unsupervised structures, such as UCL-Dehaze, demonstrated superior generalization, with PSNR 4-6 dB gains on supervised CNNs when tested on real-world data sets, like I-Haze, O-Haze, and Dense-Haze. These results emphasize the importance of domain adaptation strategies for practical implementation.

VIII. CHALLENGES AND FUTURE DIRECTIONS

Despite the remarkable advancements brought by deep learning in single image dehazing, several critical challenges persist that limit its translation into real-world applications. Drawing on the current literature and recent developments in model architecture, efficiency, and data availability, the following key challenges and promising future directions are identified as illustrated in Table 20.

A. ENHANCING GENERALIZATION ACROSS DOMAINS

A major problem with existing methods of depth haze removal is that they all rely on synthesized datasets produced using simplified scattering simulation models, which do not incorporate real-world variations such as inhomogeneous haze, non-uniform illumination, and depth information. Hence, these models train ineffectively when dealing with real-world images. As a result, these models train ineffectively when it comes to real-world images. Future work in this area can focus on the application of domain-generalization methods such as test-time adaptation, unsupervised domain transfer, and domain invariant learning to mitigate this synthetic to real-domain gap.

TABLE 19. Application-oriented impact of dehazing on downstream tasks.

Application Area	Dataset / Scenario	Downstream Task	Reported Impact of Dehazing	References
Autonomous Driving	RESIDE- β , (synthetic haze)	RTTS Object detection (Faster R-CNN, YOLO)	$\uparrow$ 12% mAP improvement with AOD-Net preprocessing; improved precision with DehazeFormer/Restormer + YOLO	[93], [123], [151], [181]
Aerial & Remote Sensing	Hazy UAV imagery, synthetic aerial datasets	Semantic segmentation, terrain classification	$\uparrow$ 8–10% IoU with GAN-based dehazing (Pix2Pix, DCPDN); improved boundary recovery using Transformers	[142], [155]
Smart Surveillance	RESIDE-RTTS (real outdoor scenes)	(real) Pedestrian detection in CCTV feeds	$\uparrow$ 9% AP with Edge-DehazeNet preprocessing; real-time enhancement with CNNs (FFA-Net, MSBDN)	[99], [116], [146]
Domain Adaptation	I-Haze, O-Haze, Dense-Haze (real-world datasets)	Generalized dehazing + downstream recognition	$\uparrow$ 4–6 dB PSNR and improved robustness using UCL-Dehaze (semi-supervised) vs. supervised CNNs	[85], [139], [148]

B. BLIND AND ZERO-SHOT DEHAZING

In practical situations, parameters like haze density and color variation in haze are not known and can change greatly. The current state-of-the-art methods make use of pre-defined prior information or labeled information, which is not very robust. The future research work can include work on blind image hazing models to infer properties from blind images, and usage of zero-shot learning to learn from blind images. Research in vision-language models can be an interesting area to work on in order to improve the model in different environments.

C. EFFICIENT AND REAL-TIME PROCESSING

To apply a dehazing algorithm on a device with limited resources, fast inference and memory efficiency are required. Although EdgeDehazeNet and FFA-Net have presented efficient methods in terms of model size, these need to be further optimized for application on devices with limited resources. Some new directions for this purpose include Mixture-of-Experts, attention pruning, low-rank approximation, neural architecture search, or model quantization.

D. ADDRESSING COMPOUND DEGRADATIONS

Real-world images are prone to a combination of degradation such as rain, snow, blur, and noise overlayed with hazes. A model solely optimized for image dehazing fails in mixed degradation situations. The future scope of study can include joint restoration methods based on multitask learning algorithms, two-branch based methods, or cross-modal learning algorithms.

E. REDUCING DEPENDENCE ON PAIRED SUPERVISION

Acquiring image pairs with hazy and clean images accurately aligned is challenging and expensive. Thus, it is important to work towards developing methods based on weakly supervised learning, self-supervised learning, contrastive learning using either unpaired or weakly labeled images. Moreover, greater emphasis should be given to evaluation metrics other than PSNR and SSIM, such as perceptually driven and semantic-aware assessment using NIQE, PI,

or task-oriented metrics such as accuracy in post-dehazing image detection tasks.

F. TRANSPARENCY AND USER INTERACTION

With the integration of dehazing into safety-critical domains such as autonomous vehicles and surveillance systems, interpretability in models acquires prominence. Future solutions can include attention visualization, uncertainty estimation, or an explanation module. User-interactive dehazing with parameters, cues, or slider inputs will increase flexibility in addition to improving user-centric control over restoration.

G. DATASET BIAS AND DOMAIN ADAPTATION

RESIDE synthetic datasets [71], with their massive scale of supervisions, do not, however, capture the intricate reality of hazy scenes. Real-world datasets including I-Haze [73], O-Haze [72], and Dense-Haze [101] successfully address the reality of a hazy image but suffer from a small size of datasets, leading to a weak capability of generalization in different domains. To overlay this bias, recent studies have started focusing on domain adaptation learning, semisupervised learning, and unsupervised learning methods, such as image adaptation using CycleGAN [108] and UCL-Dehaze [23].

H. OPEN CHALLENGES IN REAL-WORLD DEHAZING

Through the years, there have been breakthroughs in learning-based dehazing, but still the reliability of the method in such a dynamic outdoor environment remains an issue since there are several unresolved challenges associated with it.

1) NIGHT-TIME HAZE AND LOW-LIGHT CONDITIONS

Night haze brings in new difficulties such as non-uniform illumination, distortions in color due to light pollution, and increased noise artifacts. Conventional models of atmospheric scattering do not handle these illumination variations effectively and lead to over-enhanced or imbalanced color correction. Illumination-aided dehazing, which can be considered an integration of night light illumination

TABLE 20. Key challenges and proposed solutions in image dehazing.

Challenge	Description	Proposed Solutions
Domain Generalization & Open-Set Dehazing	Poor generalization from synthetic datasets to real-world haze conditions	Unsupervised domain adaptation, test-time adaptation, domain-invariant feature extraction
Blind & Zero-Shot Dehazing	Unknown haze type, density, or scene complexity during inference	Blind dehazing networks, zero-shot learning, semantic guidance via vision-language models (e.g., CLIP)
Lightweight & Real-Time Deployment	High computational cost limits edge deployment	Neural architecture search (NAS), quantization, pruning, Mixture-of-Experts (MoE), knowledge distillation
Compound Weather Degradations	Co-existence of haze with rain, snow, or motion blur	Multi-task learning, adaptive dual-path networks, multimodal fusion
Limited Supervision & Evaluation Metrics	Scarcity of paired training data and poor perceptual relevance of PSNR/SSIM	Self-/semi-supervised learning, contrastive loss, no-reference metrics (NIQE, PI), task-aware evaluation
Interpretability & User Controllability	Lack of transparency and user interaction in model predictions	Attention visualizations, confidence maps, prompt/slider-based control for user-guided dehazing
Dataset bias & Domain gap	Synthetic datasets (e.g., RESIDE) differ significantly from real-world haze, leading to poor generalization	Employ domain adaptation, semi-/unsupervised learning (e.g., CycleGAN, UCL-Dehaze), and hybrid training pipelines (synthetic pretraining + real fine-tuning).

enhancement, noise removal, and reflectance modeling, is a promising field for future research.

2) MULTI-WEATHER AND COMPOUND DEGRADATIONS (HAZE + RAIN + SNOW)

Degradations of outdoor visibility do not arise solely because of haze in most cases but have to co-occur with other complexities in the atmosphere such as rain, snow, dust, or mist. Every degradation mode has inherent distinct physical properties in an image; for instance, global reduction in image intensity in case of haze, streaking due to rain, and simulating texture and intensity reduction in case of snow. Single degradation models are not very effective when it comes to generalizing over such composite situations. Therefore, future work will have to emphasize developing composite weather restoration approaches to address such situations simultaneously.

3) UNPAIRED REAL-WORLD LEARNING AND DOMAIN ADAPTATION

The absence of corresponding real-world hazy and clean image datasets remains a major impediment to supervised methods in this area. Generalization based on purely synthetic datasets would be poor since these datasets come from a different domain compared to natural haziness. Although these challenges have been alleviated using contrastive learning and domain transfer methods based on CycleGAN translation, maintaining color information, edge information, and physical validity without paired learning remains a challenge. Some of the future directions in this field will focus on self-supervised learning methods, physics-aided learning

methods, or semi-synthetic datasets that combine simulated haziness with real-world prior statistics.

I. PHYSICS-INFORMED AND HYBRID LEARNING MODELS

Very recently, it has become evident that physics-constrained learning methods are increasingly important in single image hazing. Through a combination of the Atmospheric Scattering Model and learning methods, such physics models address overfitting issues in deep learning methods when performed on synthetic datasets. Such methods make it possible to incorporate scattering parameters, transmission prior, and atmospheric light into the learning network.

Representative examples include Physics-Guided Neural Networks (PGNNs) [181], end-to-end physical model and image translation fused network [183], and Physics-Embedded Transformers [182]. Those methods embed ASM or Enhanced ASM (EASM) equations into a transformer or CNN. Hybrid Optimization Techniques generalize this approach by mixing non-differentiable physics models with deep learning models such that a model is optimized towards both numerical and perception accuracy.

Dual path and residual hybrid approaches based on Physics-Informed DehazeNet and Domain-Constrained Diffusion Models improve reconstructing abilities by learning the residual components of hazy images using physics knowledge. Such methods are driven by physics-defined loss functions based on principles of radiative transfer.

In summary, adaptive hybrid architectures are a promising area of future research with a dynamic trade-off between physics and learning, particularly in harsh weather or spatial varying hazy environments. With the incorporation of self-supervised consistency regularization, domain

adaptation, and energy-based losses, physics-informed hybrid models are a way towards efficient, interpretable, and practical dehazing methods. To summarize these findings, a summary of challenges in image dehazing research, along with recent solutions in literature, are presented in Table 20.

IX. CONCLUSION

Single image dehazing remains a critical task for enhancing image quality in various computer vision applications, particularly in challenging outdoor and adverse weather conditions. Over the past decade, substantial progress has been made in developing dehazing methods, ranging from traditional domain-based approaches relying on physical priors to more advanced deep learning models, which have shown significant improvements in dehazing performance. These models, particularly convolutional networks, GANs, transformers, and hybrid architectures, have revolutionized the field by learning complex mappings from data, leading to better haze removal in both synthetic and real-world conditions.

A key factor in advancing dehazing techniques has been the availability and development of diverse benchmark datasets, such as RESIDE, SOTS, O-HAZE, I-HAZE, and DenseHAZE, which provide both synthetic and real-world hazy images for evaluating dehazing models. These datasets are crucial for training models and comparing performance across various dehazing techniques. However, the lack of real-world paired hazy and haze-free images remains a challenge, leading to innovations in unsupervised learning and self-supervised methods to bridge this gap.

Additionally, the design of appropriate loss functions is critical in optimizing dehazing models. While pixel-wise losses like L1 and L2 remain foundational, perceptual losses based on pre-trained neural networks, adversarial losses from GANs, and contrastive losses for self-supervised training have gained prominence. These advanced loss functions help preserve high-frequency details, enhance texture restoration, and improve perceptual quality. In particular, adversarial training has shown remarkable success in generating visually realistic outputs, while perceptual and contrastive losses enable the models to focus on semantic correctness rather than just pixel accuracy.

Metrics used to evaluate dehazing performance have evolved to better capture both pixel-level fidelity and perceptual quality. Traditional metrics like PSNR and SSIM continue to be widely used, but they are often insufficient for evaluating perceptual realism. As a result, more advanced metrics such as FSIM, GMSD, AMBE, and VSI are now commonly employed to assess the visual quality and structural integrity of restored images. These metrics provide a more holistic view of model performance, balancing both objective accuracy and subjective visual appeal.

The design of dehazing networks has evolved from early convolutional architectures to more sophisticated models incorporating multi-scale feature fusion, attention mechanisms, and transformer-based structures. Multi-scale

networks, such as MSBDN and GFN, effectively capture global and local features, while attention-based models focus on challenging regions with severe haze. Transformer models, such as Restormer and DehazeFormer, leverage self-attention mechanisms to capture long-range dependencies and enhance global consistency, leading to superior haze removal performance. Furthermore, lightweight architectures like FFA-Net and EdgeDehazeNet have been designed for real-time, low-power deployment, making them ideal for edge AI applications.

Despite these advancements, several challenges persist, including domain generalization, limited real-world paired datasets, and the high computational cost of deep learning models. To address these issues, novel strategies such as unsupervised learning, domain-invariant feature extraction, and lightweight model design have been proposed. Furthermore, real-time hardware-oriented dehazing solutions, optimized for platforms like FPGAs, DSPs, and embedded systems, have made dehazing feasible in edge AI applications. These solutions prioritize low power consumption, high processing speed, and compact memory footprints, making them suitable for resource-constrained environments.

Looking ahead, future research in image dehazing will likely focus on improving model generalization to diverse real-world conditions, reducing computational overhead, and integrating multimodal data fusion for handling complex weather scenarios. The emergence of self-supervised learning, task-aware evaluation metrics, and interpretability enhancements will further contribute to robust, real-time dehazing systems capable of performing effectively in a wide range of applications. Ultimately, the ongoing evolution of deep learning models and their deployment on specialized hardware will continue to drive significant advancements, paving the way for more accurate and efficient dehazing methods suitable for practical use in autonomous driving, surveillance, and environmental monitoring.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

AUTHOR CONTRIBUTIONS STATEMENT

Kankanala Srinivas: Conceptualization, Methodology, Supervision, and Writing—Original Draft Preparation. Nuthi Raju: Data Curation, Software Implementation, Validation, and Writing—Review and Editing. Ragini Thatikonda: Formal Analysis, Investigation, Visualization, and Writing—Review and Editing. Chintakindi Balaram Murthy: Project Administration, Resources, Funding Acquisition, and Writing—Final Review.

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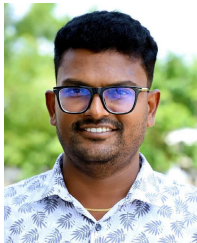
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